

Spatial Population Dynamics: Fertility, Migration, and Policies*

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Abstract

We develop a parsimonious spatial model to quantify how regional policies shape population dynamics and the distribution of income and welfare. In the model, fertility and migration are jointly determined through labor and housing markets over space. We calibrate the model using data on income, housing, migration, and fertility across 741 U.S. commuting zones (CZs), and discipline the complementarity between fertility and housing in households' preferences using an instrumental-variable-based housing–fertility moment. Counterfactual analysis yields three findings. First, reducing housing supply regulations in the 22 largest CZs to the median level across all CZs simultaneously increases aggregate fertility and per-capita welfare. Second, changes in productivity and amenities account for 36% of the variation in changes in fertility across CZs between 2000 and 2020. Third, the optimal allocation of interregional transfers depends on the welfare objective: transfers that maximize per-capita welfare concentrate workers in large CZs, exploiting agglomeration economies at the expense of population growth, whereas transfers that place greater weight on aggregate population growth retain workers in a larger number of smaller CZs, trading off per-capita real income against fertility.

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1 Introduction

Fertility has fallen sharply across advanced economies, reshaping labor supply, the age structure of populations, and the fiscal sustainability of social insurance systems. In the United States, the total fertility rate declined from 2.1 children per woman in 1990 to 1.6 in 2023, well below the replacement level. A large body of work emphasizes changes in gender roles, social norms, labor market frictions, and the difficulty of combining careers and family life as drivers of this decline (Doepke, Hannusch, Kindermann, and Tertilt, 2023; Goldin, 2021).

Yet these explanations are often silent about a central margin of fertility decisions: location. Individuals choose between large cities, which offer higher wages but also higher costs of housing and childrearing, and smaller cities, where wages are lower but housing is more affordable and space is more abundant. These trade-offs are consistent with spatial variation in fertility within the United States, as documented in Figure 1: fertility rates differ substantially across space and are lowest in the most productive and densely populated metropolitan areas.

This paper analyzes fertility as the equilibrium outcome of a spatial household problem. Households jointly choose where to live, how much housing to consume, and how many children to have, and these choices interact in general equilibrium. Spatial sorting shapes local housing demand and prices, while also affecting the cost of childrearing through the time cost reflected in wages and the space cost reflected in housing rents. In this environment, fertility responds endogenously to local economic conditions, and these responses feed back into the spatial distribution of population and economic activity.

We make three contributions. First, we develop a quantitative spatial equilibrium model that integrates endogenous fertility, location choice, and housing markets within a unified general equilibrium framework with market clearing. Relative to models that treat fertility or location in isolation, our framework allows sorting, housing prices, and demographic decisions to interact jointly in equilibrium. Second, we derive analytical expressions characterizing how local economic conditions affect fertility and how these effects propagate across regions through trade, migration, and price adjustments. These results clarify the mechanisms underlying the spatial fertility gradient and guide the quantitative policy analysis. Third, we calibrate the model to 741 U.S. commuting zones. We use an instrumental-variable-based housing–fertility moment to discipline the relevant preference elasticity and then apply the model to evaluate place-based policies, quantifying how

Figure 1: Fertility Rates and Local Economic Conditions Across U.S. Commuting Zones



Notes: Data are from the 2000 U.S. Census via IPUMS USA (Ruggles, Flood, Sobek, Backman, Cooper, Drew, Richards, Rodgers, Schroeder, and Williams, 2025). Following Autor and Dorn (2013), we use IPUMS geographic identifiers to construct 1990-consistent commuting zones. We compute the total fertility rate (TFR) for women aged 15–45 using the age of youngest child in household; average annual wages are calculated from annual wage income; following Moretti (2013), we estimate housing prices and rents at the commuting zone level. The fitted line coefficients (standard errors) in panels (a)–(d) are -0.403 (0.054), -0.091 (0.006), -0.143 (0.031), and -0.227 (0.047), respectively. Circle size represents population.

these interventions affect both aggregate efficiency and aggregate fertility through their general equilibrium effects.

Our model builds on the quantitative spatial economics framework of Redding and Rossi-Hansberg (2017), extending it to incorporate overlapping generations with endogenous fertility following Barro and Becker (1989). The economy consists of multiple regions, each producing a differentiated variety of goods. Workers are mobile across regions and face idiosyncratic location

preferences, generating the smooth migration responses characteristic of discrete choice models (McFadden, 1973). Housing is supplied with region-specific elasticities, capturing the heterogeneous regulatory and geographic constraints documented by Saiz (2010).

The key innovation is that households jointly choose where to live, how much to consume (in goods and housing), and how many children to have. Children require space, so we allow for the possibility that housing and fertility are complements in preferences, implying that the cost of childrearing may rise with housing rents. Children also require parental time, generating an opportunity cost that increases with wages. As a result, fertility varies systematically with both wages and housing rents across space.

The model yields several key insights about the mechanisms linking local economic conditions to fertility and aggregate outcomes. First, we derive closed-form expressions for the local price elasticities of fertility. When housing and children are complements, as implied by our calibration, the rent elasticity of fertility is unambiguously negative. Higher wages have an ambiguous effect, combining a negative substitution effect (children become relatively expensive) with a positive income effect. Under our calibrated parameters, the substitution effect dominates, rationalizing the negative fertility-wage correlation in the data. Second, we characterize how productivity shocks propagate through the spatial economy. A productivity boom in one region raises local wages and attracts migrants, bidding up housing costs and reducing fertility. But the shock also affects other regions: trading partners benefit from lower import prices, while labor market competitors lose workers and see rents decline. These cross-place spillovers mean that local policies have national consequences. Third, we show that endogenous fertility acts as a stabilizing force in spatial dynamics. Regions receiving population inflows experience rising rents and declining fertility, providing an equilibrating mechanism that moderates population growth beyond migration alone.

To discipline the magnitude of the housing–fertility relationship, we construct a reduced-form moment from a panel of U.S. commuting zones. We regress the regional total fertility rate on the average number of bedrooms in the region and interact bedrooms with the local homeownership share, allowing the estimated relationship to vary with the prevalence of homeownership (Dettling and Kearney, 2014). Housing space is instrumented using the interaction between national housing price shocks and local geographic constraints on housing supply. During national housing booms, house prices tend to rise more in regions with tighter geographic constraints on new construction, generating differential changes in the housing space available to local households. Conditional on

commuting-zone and year fixed effects and the included controls, this design provides an IV-based source of variation in housing space. At the reference homeownership share used to construct the calibration target, our estimates imply that a 0.1-bedroom increase is associated with an average increase of about 5 percent in fertility.

We calibrate the model to match data on wages, population, fertility, housing rents, migration flows, and trade flows across 741 U.S. commuting zones. In particular, the elasticity of substitution between housing and fertility is jointly disciplined by the IV-based housing–fertility moment described above and the other calibration targets. The resulting calibrated elasticity is 0.38, indicating strong complementarity between housing and fertility in households’ preferences. Under this calibration, policies that reduce housing costs raise fertility substantially, whereas policies that attract workers to expensive cities may reduce aggregate fertility.

The calibrated model delivers a close fit to the data. When we allow the parenting costs to vary across regions, the model exactly matches observed fertility rates. Under uniform parenting cost, the model (i) accounts for 16% of the cross-sectional variation in fertility through the endogenous responses to wages and housing costs and (ii) reproduces the negative correlations between fertility and wages, population, and housing rents documented in Figure 1. The model also matches housing rents closely, with an R^2 of 0.63. These results suggest that the model captures key forces shaping the spatial distribution of fertility and housing costs.

Armed with the calibrated model, we quantify the effects of regional policies, accounting for endogenous migration and fertility decisions, through the following exercises.

First, we quantify the effects of housing supply reform. Specifically, we relax housing supply regulations in the 22 largest CZs to the median level across all CZs. The reform expands housing supply in large, productive regions where housing markets are most constrained. As a result, it raises the balanced-growth-path (BGP) population growth rate from 0.9188 to 0.9281 and increases per-capita welfare by 1.83%. Unlike policies that shift households toward low-fertility large cities, relaxing housing supply regulations in these productive cities promotes both aggregate productivity and fertility.

Second, we quantify the contribution of changes in productivity, amenities, and parenting costs to changes in fertility rates across CZs between 2000 and 2020. We find that changes in productivity and amenities, operating through wages and housing prices, account for 36% of the variation

in changes in fertility rates over this period. This result suggests that the spatial household problem characterized in our model is quantitatively important in explaining fertility dynamics across regions and over time.

Third, we characterize interregional transfers that maximize the weighted average of per-capita welfare and aggregate population growth. Transfers that put more weight on per-capita welfare tend to concentrate workers in a small number of large CZs, exploiting agglomeration effects at the expense of fertility. As a result, they substantially raise per-capita real income while reducing aggregate population growth. By contrast, transfers that place greater weight on population growth retain workers in a larger number of small CZs, balancing per-capita real income and population growth. These findings indicate that interregional transfers can have meaningful effects on fertility and, in turn, population dynamics, with their optimal design depending on the policy objective.

Related Literature

Our paper contributes to several strands of the literature. First, we build on the quantitative spatial economics literature that studies the distribution of economic activity across space ([Redding and Rossi-Hansberg, 2017](#); [Allen and Arkolakis, 2014](#); [Caliendo, Parro, Rossi-Hansberg, and Sarte, 2018](#)). This literature has examined how trade costs, migration frictions, and local amenities shape the spatial equilibrium, with applications to regional policy ([Fajgelbaum, Morales, Suárez Serrato, and Zidar, 2019](#); [Fajgelbaum and Gaubert, 2020](#)), infrastructure investment ([Allen and Arkolakis, 2014](#)), and the gains from labor mobility ([Bryan and Morten, 2019a](#); [Tombe and Zhu, 2019](#)). A key finding is that spatial misallocation reduces aggregate output substantially ([Hsieh and Moretti, 2019](#)). Our contribution is to incorporate endogenous fertility into this framework, showing that the demographic consequences of spatial reallocation can be quantitatively important and may partially offset the efficiency gains from labor mobility.

Second, we contribute to the literature on the economics of fertility. The foundational work of [Becker \(1960\)](#) and [Becker and Lewis \(1973\)](#) established that fertility responds to the opportunity cost of children, with the quantity-quality tradeoff playing a central role. [Barro and Becker \(1989\)](#) embedded this choice into a dynastic growth model. More recent work has examined how housing costs affect fertility decisions ([Dettling and Kearney, 2014](#)), how labor market conditions shape family formation ([Autor, Dorn, and Hanson, 2019](#)), and how family policies influence demographic

outcomes (Zhou, 2025). A growing reduced-form literature documents that house-price increases raise fertility for homeowners through wealth effects, while effects for renters are small or negative (Detting and Kearney, 2014; Lovenheim and Mumford, 2013). Our contribution is to embed these mechanisms in a spatial general equilibrium model where housing prices respond endogenously to sorting. The observed negative correlation between prices and fertility thus conflates compositional sorting of low-fertility households into expensive locations with behavioral responses to prices. Our framework provides a way to consider these channels within a common general equilibrium setting.

Third, we connect to a growing literature on the interaction between economic geography and demography. Sato and Yamamoto (2005) and Sato (2007) develop theoretical models linking urbanization, migration, and fertility. Goto and Minamimura (2019) analyze the interaction between agglomeration economies and endogenous fertility in a two-region model. Most closely related is the concurrent paper by Arkolakis, Kwak, Woo, and Yang (2026), who develop a dynamic spatial equilibrium model with endogenous fertility calibrated to South Korea. Their paper emphasizes the quantity-quality tradeoff and education competition as drivers of low fertility in dense urban areas, showing that high-skilled parents sort into Seoul for higher wages, which intensifies education competition and raises the marginal cost of child quality. Our paper complements theirs in several respects. First, while Arkolakis et al. (2026) model fertility as a discrete choice with quality-quantity tradeoffs shaped by local education competition, we adopt a continuous fertility choice within a nested CES framework emphasizing the complementarity between housing and children. This specification yields closed-form expressions for fertility elasticities and transparently characterizes how the degree of substitutability shapes policy responses. Second, our papers target different empirical settings: they exploit rich Korean administrative data to study an extreme low-fertility environment, while we calibrate to U.S. commuting zones where fertility variation is less extreme but spatial misallocation concerns are prominent. Third, we emphasize different policy margins. While Arkolakis et al. (2026) focus on education cost reduction and find that a nationwide reduction equivalent to 20 percent of consumption raises fertility by 23 percent, we examine housing supply reforms and find that equalizing supply elasticities raises the population growth rate by 27 percent. Both papers reach a common conclusion: place-based policies can have large effects on aggregate fertility when they address the structural sources of low fertility in productive locations.

Fourth, our paper relates to the literature on housing supply and its macroeconomic consequences. [Saiz \(2010\)](#) provides estimates of housing supply elasticities across U.S. metropolitan areas. [Hsieh and Moretti \(2019\)](#) show that housing supply constraints in productive cities reduce aggregate output by preventing workers from accessing high-productivity locations. We extend this analysis by showing that housing constraints also affect fertility: supply-constrained cities have higher rents, which reduces family formation. This demographic channel provides an additional margin through which housing policy affects aggregate outcomes, one that operates over longer horizons than the static efficiency effects emphasized in the existing literature.

Finally, our work contributes to the literature on place-based policies. [Fajgelbaum et al. \(2019\)](#) develop a framework for evaluating state tax policies, showing that tax competition leads to spatial misallocation. [Fajgelbaum and Gaubert \(2020\)](#) characterize optimal spatial policies in the presence of agglomeration externalities. Our contribution is to introduce fertility as a policy-relevant outcome. Place-based policies have demographic consequences that should be considered alongside their effects on efficiency and equity. Our counterfactual analysis shows that housing supply reform can simultaneously improve allocative efficiency and raise aggregate fertility, suggesting that the efficiency-demography tradeoff need not bind for well-designed policies.

The remainder of the paper proceeds as follows. Section 2 presents the model and characterizes the equilibrium. Section 3 develops analytical results on the mechanisms through which local conditions affect fertility and how these effects propagate through the spatial economy. Section 4 describes the calibration strategy and data. Section 5 presents counterfactual policy exercises. Section 6 concludes.

2 Model

We develop an overlapping generations model of spatial equilibrium with endogenous fertility, migration, and housing markets. Our framework builds on the quantitative spatial economics literature ([Redding and Rossi-Hansberg, 2017](#)), extending it to incorporate demographic dynamics. The model captures how local economic conditions—particularly housing costs and wages—shape fertility decisions, and how these demographic responses feed back into the spatial distribution of economic activity.

2.1 Environment

Demographics and Geography. Time is discrete and indexed by $t = 0, 1, 2, \dots$. The economy consists of N regions. Each agent lives for two periods: young and adult. During the young period, an agent resides with her parent in birth region o and makes no economic decisions. At the beginning of the adult period, she observes idiosyncratic preference shocks for all locations and chooses whether to remain in her birth region or migrate to another region i . In her destination, she supplies labor, consumes goods and housing, and chooses how many children to have. At the end of the adult period, she dies, and her children enter adulthood in the following period. This two-period OLG structure follows [Samuelson \(1958\)](#) and provides a tractable framework for studying intergenerational population dynamics.

Preferences. We assume that young agents do not consume. In period t , an adult agent v who moves from region o to region i has utility

$$U_{oi,t}(v) = \frac{B_{i,t}(v)}{d_{oi,t}} V_{i,t}(v), \quad (1)$$

$$V_{i,t}(v) \equiv \left[\alpha^{\frac{1}{\gamma}} C_{i,t}(v)^{\frac{\gamma-1}{\gamma}} + (1-\alpha)^{\frac{1}{\gamma}} \left[\beta^{\frac{1}{\rho}} h_{i,t}(v)^{\frac{\rho-1}{\rho}} + (1-\beta)^{\frac{1}{\rho}} n_{i,t}(v)^{\frac{\rho-1}{\rho}} \right]^{\frac{\rho}{\rho-1} \frac{\gamma-1}{\gamma}} \right]^{\frac{\gamma}{\gamma-1}},$$

where $C_{i,t}(v)$ denotes consumption of a composite good, $h_{i,t}(v)$ denotes housing services, and $n_{i,t}(v)$ denotes the number of children. The term $d_{oi,t} \geq 1$ represents the iceberg migration friction between regions o and i , with $d_{oo,t} = 1$ for stayers.

The preference specification embeds a nested CES structure that governs substitution patterns across the three arguments of the utility function. The outer nest, governed by parameter $\gamma \geq 0$, determines the elasticity of substitution between consumption goods and a composite of housing and children. The inner nest, governed by $\rho \geq 0$, determines the elasticity of substitution between housing and children. The parameters $\alpha \in (0, 1)$ and $\beta \in (0, 1)$ control expenditure shares. This nested structure allows for flexible substitution patterns: when housing becomes expensive, households may substitute toward fewer children (if $\rho > 1$) or toward consumption goods (if $\gamma > 1$). The interaction between these margins is central to our analysis of how housing costs affect fertility.

The term $B_{i,t}(v)$ captures idiosyncratic taste shocks for location i , which generate dispersion

in location choices even among otherwise identical agents. Following [McFadden \(1973\)](#) and the subsequent spatial economics literature ([Eaton and Kortum, 2002](#); [Bryan and Morten, 2019b](#); [Redding, 2016](#)), we assume these shocks are drawn from a Fréchet distribution:

$$\Pr[B_{i,t}(v) \leq b] = \exp[-B_{i,t}b^{-\eta}], \quad (2)$$

where $B_{i,t} > 0$ is a location-specific amenity shifter and $\eta > 1$ governs the dispersion of idiosyncratic tastes. A higher η implies less dispersion, making agents more responsive to systematic differences in utility across locations.

Production. Each region produces a distinct variety of goods under perfect competition. Varieties are aggregated through a CES consumption index with elasticity of substitution $\sigma > 1$, following [Dixit and Stiglitz \(1977\)](#) and [Krugman \(1980\)](#). The unit cost of producing variety i is

$$uc_{i,t} = \frac{w_{i,t}}{A_{i,t} \left(L_{i,t}^e \right)^\psi}, \quad (3)$$

where $w_{i,t}$ denotes the wage, $A_{i,t}$ is an exogenous productivity shifter, $L_{i,t}^e$ is effective labor in region i , and $\psi \geq 0$ captures external economies of scale. The agglomeration externality reflects productivity spillovers from the concentration of economic activity, as emphasized by [Ciccone and Hall \(1996\)](#) and [Combes, Duranton, Gobillon, Puga, and Roux \(2012\)](#). All varieties are tradable, subject to iceberg trade costs $\tau_{in,t} \geq 1$ from region i to region n , with $\tau_{ii,t} = 1$.

Housing Supply. Following [Saiz \(2010\)](#) and [Hsieh and Moretti \(2019\)](#), the inverse supply function of housing in region i is

$$r_{i,t} = \bar{r}_{i,t} H_{i,t}^{\xi_i}, \quad (4)$$

where $\xi_i \geq 0$ is the inverse elasticity of housing supply, $H_{i,t}$ is total housing supply, and $\bar{r}_{i,t}$ is an exogenous shifter. A higher ξ_i implies more inelastic supply: as housing demand increases, rents rise more sharply in supply-constrained regions.

Household Problem. The initial distribution of adults, $L_{i,0}$, is given exogenously. Each adult is endowed with one unit of time, which she allocates between market work and parenting. Raising

each child requires $\chi_{i,t}$ units of time, capturing the opportunity cost of fertility in terms of forgone labor income. This formulation follows [Becker \(1960\)](#) and [Becker and Lewis \(1973\)](#).

Conditional on moving to region i from region o in period t , the adult solves:

$$V_{i,t} = \max_{\{C_{i,t}, h_{i,t}, n_{i,t}\}} \left[\alpha^{\frac{1}{\gamma}} C_{i,t}^{\frac{\gamma-1}{\gamma}} + (1-\alpha)^{\frac{1}{\gamma}} \left[\beta^{\frac{1}{\rho}} h_{i,t}^{\frac{\rho-1}{\rho}} + (1-\beta)^{\frac{1}{\rho}} n_{i,t}^{\frac{\rho-1}{\rho}} \right]^{\frac{\rho}{\rho-1} \frac{\gamma-1}{\gamma}} \right]^{\frac{\gamma}{\gamma-1}} \quad (5)$$

s.t. $P_{i,t}C_{i,t} + r_{i,t}h_{i,t} = w_{i,t}(1 - \chi_{i,t}n_{i,t}) + T_{i,t},$

where $P_{i,t}$ is the price index of the consumption good composite, $r_{i,t}$ is the housing rent, and $T_{i,t}$ is a lump-sum transfer. The right-hand side of the budget constraint reflects the opportunity cost structure of fertility: each child reduces labor income by $w_{i,t}\chi_{i,t}$. We further assume that

$$\chi_{i,t} = \bar{\chi}_{i,t} L_{i,t}^{\zeta}, \quad (6)$$

where ζ reflects the extent to which parenting costs vary with regional population, with $\zeta < 0$ suggesting a scale economy of parenting in larger regions and with $\zeta > 0$ suggesting a fertility-specific congestion force.

Housing Revenue Distribution. In our baseline specification, these revenues are evenly distributed to all households regardless of location. Let $L_{i,t}$ denote total adult population

$$T_{i,t} = T_t \equiv \frac{\sum_{i=1}^N r_{i,t} H_{i,t}}{\sum_{i=1}^N L_{i,t}}. \quad (7)$$

This assumption ensures that the distribution of housing rents does not directly affect location choices through wealth effects.

2.2 Equilibrium

We now characterize the equilibrium. Let $L_{oi,t}$ denote the number of adults born in region o who live in region i in period t :

$$L_{i,t} \equiv \sum_{o=1}^N L_{oi,t}. \quad (8)$$

Effective labor supply equals population minus time devoted to childrearing:

$$L_{i,t}^e = (1 - \chi_{i,t} n_{i,t}) L_{i,t}. \quad (9)$$

This expression highlights a key margin: higher fertility reduces effective labor supply, affecting local productivity through agglomeration and, consequently, wages and prices throughout the economy.

Optimal Consumption and Fertility. The household's optimization yields demand functions for goods, housing, and children. Define the expenditure index:

$$\Xi_{i,t} \equiv \alpha P_{i,t}^{1-\gamma} + (1 - \alpha) \left[\beta r_{i,t}^{1-\rho} + (1 - \beta) (\chi_{i,t} w_{i,t})^{1-\rho} \right]^{\frac{1-\gamma}{1-\rho}}. \quad (10)$$

Optimal goods consumption is:

$$C_{i,t} = \frac{1}{P_{i,t}} \frac{\alpha P_{i,t}^{1-\gamma}}{\Xi_{i,t}} (w_{i,t} + T_t). \quad (11)$$

Optimal housing consumption is:

$$h_{i,t} = \frac{1}{r_{i,t}} \frac{\beta r_{i,t}^{1-\rho}}{\beta r_{i,t}^{1-\rho} + (1 - \beta) (\chi_{i,t} w_{i,t})^{1-\rho}} \frac{(1 - \alpha) \left[\beta r_{i,t}^{1-\rho} + (1 - \beta) (\chi_{i,t} w_{i,t})^{1-\rho} \right]^{\frac{1-\gamma}{1-\rho}}}{\Xi_{i,t}} (w_{i,t} + T_t). \quad (12)$$

Optimal fertility is:

$$n_{i,t} = \frac{1}{\chi_{i,t} w_{i,t}} \frac{(1 - \beta) (\chi_{i,t} w_{i,t})^{1-\rho}}{\beta r_{i,t}^{1-\rho} + (1 - \beta) (\chi_{i,t} w_{i,t})^{1-\rho}} \frac{(1 - \alpha) \left[\beta r_{i,t}^{1-\rho} + (1 - \beta) (\chi_{i,t} w_{i,t})^{1-\rho} \right]^{\frac{1-\gamma}{1-\rho}}}{\Xi_{i,t}} (w_{i,t} + T_t). \quad (13)$$

These expressions reveal the key economic forces shaping fertility. The fertility rate depends on the ratio of housing rents to the opportunity cost of children, $r_{i,t}/(\chi_{i,t} w_{i,t})$, through the inner CES aggregator. When housing is expensive relative to the time cost of children, households substitute toward fewer children and more housing (if $\rho > 1$). Income raises fertility through an income

effect, while higher wages also raise the opportunity cost of children, generating a substitution effect working in the opposite direction.

Trade and Prices. Goods trade follows the standard gravity structure. The share of region n 's expenditure on goods from region i is:

$$\lambda_{in,t} \equiv \frac{X_{in,t}}{X_{n,t}} = (\tau_{in,t} w_{i,t})^{1-\sigma} \left(A_{i,t} (L_{i,t}^e)^\psi \right)^{\sigma-1} P_{n,t}^{\sigma-1}, \quad (14)$$

where the price index in region n is:

$$P_{n,t} = \left[\sum_{i=1}^N (\tau_{in,t} w_{i,t})^{1-\sigma} \left(A_{i,t} (L_{i,t}^e)^\psi \right)^{\sigma-1} \right]^{\frac{1}{1-\sigma}}. \quad (15)$$

Market Clearing. Wages are determined by labor market clearing:

$$w_{i,t} L_{i,t}^e = \sum_{n=1}^N \lambda_{in,t} X_{n,t}. \quad (16)$$

Total goods expenditure equals consumption expenditure:

$$X_{i,t} = P_{i,t} C_{i,t} L_{i,t} = \left[\frac{\alpha P_{i,t}^{1-\gamma}}{\Xi_{i,t}} (w_{i,t} + T_t) \right] L_{i,t}. \quad (17)$$

Housing market clearing requires:

$$\begin{aligned} H_{i,t} &= h_{i,t} L_{i,t} \\ &= \frac{1}{r_{i,t}} \left[\frac{\beta r_{i,t}^{1-\rho}}{\beta r_{i,t}^{1-\rho} + (1-\beta)(\chi_{i,t} w_{i,t})^{1-\rho}} \frac{(1-\alpha) \left[\beta r_{i,t}^{1-\rho} + (1-\beta)(\chi_{i,t} w_{i,t})^{1-\rho} \right]^{\frac{1-\gamma}{1-\rho}}}{\Xi_{i,t}} (w_{i,t} + T_t) \right] L_{i,t}. \end{aligned} \quad (18)$$

Migration and Population Dynamics. The number of young agents in region o at the end of period $t-1$ equals $n_{o,t-1} L_{o,t-1}$. These agents become adults in period t and choose where to live.

Given the Fréchet distribution, the share of adults born in o who migrate to i is:

$$\frac{L_{oi,t}}{n_{o,t-1}L_{o,t-1}} = \frac{B_{i,t} \left(\frac{1}{d_{oi,t}} V_{i,t} \right)^\eta}{\sum_{k=1}^N B_{k,t} \left(\frac{1}{d_{ok,t}} V_{k,t} \right)^\eta}, \quad (19)$$

where utility is:

$$V_{i,t} = \left[\alpha^{\frac{1}{\gamma}} C_{i,t}^{\frac{\gamma-1}{\gamma}} + (1-\alpha)^{\frac{1}{\gamma}} \left[\beta^{\frac{1}{\rho}} h_{i,t}^{\frac{\rho-1}{\rho}} + (1-\beta)^{\frac{1}{\rho}} n_{i,t}^{\frac{\rho-1}{\rho}} \right]^{\frac{\rho}{\rho-1} \frac{\gamma-1}{\gamma}} \right]^{\frac{\gamma}{\gamma-1}}. \quad (20)$$

This migration equation embodies the core spatial equilibrium logic: regions with higher utility attract more migrants. The presence of fertility in both the numerator of (19)—through the origin region's fertility rate $n_{o,t-1}$ —and in the destination's attractiveness $V_{i,t}$ creates a rich dynamic interaction between demographic and spatial forces.

Equilibrium Definition.

Definition 1 (Equilibrium). *Given exogenous fundamentals $(A_{i,t}, B_{i,t}, d_{oi,t}, \tau_{in,t}, \chi_{i,t}, \bar{r}_{i,t})$, parameters $(\eta, \psi, \sigma, \gamma, \rho, \alpha, \beta, \xi_i)$, and initial conditions $(L_{i,0})$, a dynamic equilibrium is a sequence of prices $(w_{i,t}, r_{i,t}, P_{i,t})$, allocations $(n_{i,t}, H_{i,t})$, and population distributions $(L_{i,t})$ such that for all $t \geq 0$:*

1. *Wages $w_{i,t}$ clear the labor market according to Equation (16).*
2. *Housing rents $r_{i,t}$ clear the housing market according to Equation (18).*
3. *Housing supply $H_{i,t}$ satisfies Equation (4).*
4. *Price indices $P_{i,t}$ are given by Equation (15).*
5. *Fertility rates $n_{i,t}$ satisfy Equation (13).*
6. *Population $L_{i,t}$ evolves according to Equations (19) and (8).*

Expected Welfare. The Fréchet structure yields a convenient expression for expected welfare. The expected utility of an agent born in region o who enters adulthood in period t is:

$$W_{o,t} = \Gamma\left(1 - \frac{1}{\eta}\right) \left[\sum_{k=1}^N B_{k,t} \left(\frac{1}{d_{ok,t}} V_{k,t} \right)^\eta \right]^{\frac{1}{\eta}}, \quad (21)$$

where $\Gamma(\cdot)$ is the gamma function. This welfare measure aggregates opportunities across all potential destinations and forms the basis for our welfare analysis.

2.3 Steady State

To characterize long-run properties, we analyze steady states in which population shares remain constant over time.

Definition 2 (Steady State). *A steady state is an equilibrium in which population in all regions grows at a common constant rate:*

$$L_{i,t} = g L_{i,t-1}, \quad g > 0, \quad \forall i, t. \quad (22)$$

In steady state, the growth rate g equals the population-weighted average fertility rate:

$$g = \frac{\sum_{i=1}^N n_{i,t} L_{i,t}}{\sum_{i=1}^N L_{i,t}}, \quad \forall t. \quad (23)$$

This condition ensures demographic consistency: the number of children born each period equals the number of new adults. If $g > 1$, the population grows; if $g < 1$, it declines. The steady-state growth rate is determined endogenously by the interaction of local fertility decisions, migration patterns, and economic conditions.

We drop time subscripts when expressing steady-state quantities and denote detrended population as $L_i \equiv L_{i,t} g^{-t}$.

Definition 3 (Steady-State Equilibrium). *Given fundamentals $(A_i, B_i, d_{oi}, \tau_{in}, \chi_i, \bar{r}_i)$ and parameters $(\eta, \psi, \sigma, \gamma, \rho, \alpha, \beta, \xi_i)$, a steady-state equilibrium consists of prices (w_i, r_i, P_i) , allocations (n_i, H_i) , population (L_i) , and growth rate g such that:*

1. **Labor market clearing:** Wages (w_i) satisfy

$$w_i L_i^e = \sum_{n=1}^N \lambda_{in} P_n C_n L_n, \quad \lambda_{in} = (\tau_{in} w_i)^{1-\sigma} (A_i (L_i^e)^\psi)^{\sigma-1} P_n^{\sigma-1}, \quad L_i^e = (1 - \chi_i n_i) L_i, \quad C_n = \frac{1}{P_n} \frac{\alpha P_n^{1-\gamma}}{\Xi_n} (w_n + T_n), \quad (24)$$

where the transfer is

$$T_i = T = \frac{\sum_{k=1}^N r_k H_k}{\sum_{k=1}^N L_k}, \quad (25)$$

the expenditure index is

$$\Xi_i \equiv \alpha P_i^{1-\gamma} + (1 - \alpha) \left[\beta r_i^{1-\rho} + (1 - \beta) (\chi_i w_i)^{1-\rho} \right]^{\frac{1-\gamma}{1-\rho}}, \quad (26)$$

and the parenting cost is

$$\chi_i = \bar{\chi}_i L_i^\zeta. \quad (27)$$

2. **Housing market clearing:** Rents (r_i) and housing supply (H_i) satisfy

$$r_i = \bar{r}_i H_i^{\xi_i}, \quad H_i = \frac{1}{r_i} \left[\frac{\beta r_i^{1-\rho}}{\beta r_i^{1-\rho} + (1 - \beta) (\chi_i w_i)^{1-\rho}} \frac{(1 - \alpha) \left[\beta r_i^{1-\rho} + (1 - \beta) (\chi_i w_i)^{1-\rho} \right]^{\frac{1-\gamma}{1-\rho}}}{\Xi_i} (w_i + T_i) \right] L_i. \quad (28)$$

3. **Price indices:** (P_i) satisfy

$$P_n = \left[\sum_{i=1}^N (\tau_{in} w_i)^{1-\sigma} (A_i (L_i^e)^\psi)^{\sigma-1} \right]^{\frac{1}{1-\sigma}}. \quad (29)$$

4. **Fertility:** Fertility rates (n_i) satisfy

$$n_i = \frac{1}{\chi_i w_i} \frac{(1 - \beta) (\chi_i w_i)^{1-\rho}}{\beta r_i^{1-\rho} + (1 - \beta) (\chi_i w_i)^{1-\rho}} \frac{(1 - \alpha) \left[\beta r_i^{1-\rho} + (1 - \beta) (\chi_i w_i)^{1-\rho} \right]^{\frac{1-\gamma}{1-\rho}}}{\Xi_i} (w_i + T_i). \quad (30)$$

5. **Population distribution:** The detrended population (L_i) satisfies

$$g L_i = \sum_{o=1}^N \frac{B_i \left(\frac{1}{d_{oi}} V_i \right)^\eta}{\sum_{k=1}^N B_k \left(\frac{1}{d_{ok}} V_k \right)^\eta} n_o L_o, \quad (31)$$

$$\text{where } V_i = \left[\alpha^{\frac{1}{\gamma}} C_i^{\frac{\gamma-1}{\gamma}} + (1-\alpha)^{\frac{1}{\gamma}} \left[\beta^{\frac{1}{\rho}} h_i^{\frac{\rho-1}{\rho}} + (1-\beta)^{\frac{1}{\rho}} n_i^{\frac{\rho-1}{\rho}} \right]^{\frac{\rho}{\rho-1}} \right]^{\frac{\gamma}{\gamma-1}}.$$

6. **Aggregate fertility:** The growth rate g satisfies

$$g = \frac{\sum_{i=1}^N n_i L_i}{\sum_{i=1}^N L_i}. \quad (32)$$

Several features of this equilibrium merit emphasis. First, the population growth rate g is endogenous and depends on the distribution of economic activity across regions with different fertility rates. Policies that reallocate population toward high-fertility regions raise aggregate fertility and the long-run growth rate. Second, the interaction between housing supply elasticities and fertility creates spatial variation in demographic responses to economic shocks. Third, migration links regions together: a shock to one region affects fertility and welfare throughout the economy through trade, migration, and price adjustments.

The steady-state expected utility in region o is:

$$W_o = \Gamma \left(1 - \frac{1}{\eta} \right) \left[\sum_{k=1}^N B_k \left(\frac{1}{d_{ok}} V_k \right)^{\eta} \right]^{\frac{1}{\eta}}. \quad (33)$$

The aggregate welfare in the steady state is therefore

$$W \equiv \sum_{o=1}^N \iota_o W_o, \quad \iota_o \equiv \frac{L_o}{\sum_{k=1}^N L_k}. \quad (34)$$

3 Model Mechanisms

This section develops analytical results that illuminate the model's key mechanisms. To this end, we analyze the propagation of local shocks through the spatial economy, highlighting the role of endogenous fertility in shaping both local and aggregate responses.

3.1 Local Determinants of Fertility

We begin by characterizing how fertility responds to local economic conditions. Define the expenditure shares on consumption goods, housing, and children as:

$$\omega_i^C \equiv \frac{P_i C_i}{w_i + T_i}, \quad \omega_i^h \equiv \frac{r_i h_i}{w_i + T_i}, \quad \omega_i^n \equiv \frac{\chi_i w_i n_i}{w_i + T_i}, \quad (35)$$

and the within-nest shares as:

$$\tilde{\omega}_i^h \equiv \frac{\beta r_i^{1-\rho}}{\beta r_i^{1-\rho} + (1-\beta)(\chi_i w_i)^{1-\rho}}, \quad \tilde{\omega}_i^n \equiv 1 - \tilde{\omega}_i^h. \quad (36)$$

Proposition 1 (Local Fertility Elasticities). *Holding transfers T_i fixed, the elasticities of aggregate local fertility with respect to wages, rents, and the time cost of children are:*

$$\varepsilon_{n,w} \equiv \frac{\partial \log n_i}{\partial \log w_i} \Big|_{T_i} = \underbrace{-\rho - (\gamma - \rho)\tilde{\omega}_i^n - (1 - \gamma)(1 - \omega_i^C)\tilde{\omega}_i^n}_{\text{substitution effect}} + \underbrace{\frac{w_i}{w_i + T_i}}_{\text{income effect}} \quad (37)$$

$$\varepsilon_{n,r} \equiv \frac{\partial \log n_i}{\partial \log r_i} \Big|_{T_i} = \underbrace{\tilde{\omega}_i^h [-(\gamma - \rho) - (1 - \gamma)(1 - \omega_i^C)]}_{\text{price effect}} \quad (38)$$

$$\varepsilon_{n,\chi} \equiv \frac{\partial \log n_i}{\partial \log \chi_i} = \varepsilon_{n,w} - \frac{w_i}{w_i + T_i} \quad (39)$$

The rent elasticity $\varepsilon_{n,r}$ is non-positive if and only if $-(1 - \gamma)(1 - \omega_i^C) \leq \gamma - \rho$. This condition holds whenever $\rho \leq \gamma \leq 1$. When transfers respond endogenously to rents, the total rent elasticity includes an additional wealth effect:

$$\frac{\partial \log n_i}{\partial \log r_i} = \varepsilon_{n,r} + \frac{T_i}{w_i + T_i} \cdot \varepsilon_{T,r}, \quad (40)$$

where $\varepsilon_{T,r} \equiv \partial \log T_i / \partial \log r_i > 0$.

Proof. See Appendix A.1. □

These elasticities admit clear economic interpretations. Consider first the wage elasticity $\varepsilon_{n,w}$. When wages rise, the opportunity cost of children $\chi_i w_i$ increases, making children relatively more

expensive. This generates a *substitution effect* with three components. First, the direct effect $-\rho$ captures the own-price elasticity of fertility demand within the inner nest, holding composite prices fixed. Second, the within-nest feedback $-(\gamma - \rho)\tilde{\omega}_i^n$ reflects how the higher child cost raises the housing-fertility composite price, which in turn affects fertility demand. When $\gamma > \rho$, the outer nest is more substitutable than the inner nest, amplifying the response; when $\gamma < \rho$, it dampens the response. Third, the cross-nest substitution $-(1 - \gamma)(1 - \omega_i^C)\tilde{\omega}_i^n$ captures reallocation toward consumption goods. When $\gamma < 1$, consumption and the housing-fertility composite are complements, so the higher composite price reduces demand for both.

Working against the substitution effect is the *income effect*: higher wages expand the budget set, and since children are a normal good, fertility rises with income. The income effect is captured by $w_i/(w_i + T_i)$, the share of income derived from wages. The net wage elasticity is negative when the substitution effect dominates.

The rent elasticity $\varepsilon_{n,r}$ captures how housing costs affect fertility through the *price effect*. Higher rents raise the price of housing within the inner nest and thereby increase the price index of the housing-fertility composite. The structure parallels the wage elasticity but without the direct $-\rho$ term, since rents do not directly enter the opportunity cost of children. The overall effect is negative when $\rho \leq \gamma < 1$.

The time cost elasticity $\varepsilon_{n,\chi}$ equals the substitution component of the wage elasticity. The relationship $\varepsilon_{n,\chi} = \varepsilon_{n,w} - w_i/(w_i + T_i)$ follows because χ_i and w_i enter the opportunity cost multiplicatively: both raise the effective price $\chi_i w_i$, but only wages generate an income effect. This relationship implies that the time cost elasticity is always more negative than the wage elasticity.

In sum, the local elasticities in Proposition 1 indicate that:

1. *Higher wages reduce fertility* when the substitution effect—comprising the direct price effect, within-nest feedback, and cross-nest substitution toward consumption—dominates the income effect. The calibrated complementarities ($\rho < 1$ and $\gamma < 1$) ensure this condition holds.
2. *Higher housing costs reduce fertility* through the price effect when $\gamma < 1$ and consumption is not too large a share of expenditure. This negative effect may be partially offset by a positive wealth effect when housing revenues are distributed to local residents.

3.2 General Equilibrium Propagation

Local shocks propagate through the spatial economy via goods trade, worker migration, and price adjustment. Understanding these channels is essential for policy analysis, since interventions in one region affect outcomes elsewhere.

Proposition 2 (Spatial Propagation of Shocks). *Consider a productivity shock $\hat{A}_i > 0$ in region j , where $\hat{x} \equiv d \log x$ denotes log-deviations. The shock propagates through three channels:*

Trade channel. *Holding wages and employment fixed, region i 's price index responds to productivity in region j according to:*

$$\left. \frac{\partial \log P_n}{\partial \log A_i} \right|_{w, L^e} = -\lambda_{in} < 0, \quad (41)$$

where λ_{in} is region n 's expenditure share on goods from i . In general equilibrium, wages and employment adjust, yielding:

$$\hat{P}_n = \sum_{i=1}^N \lambda_{in} (\hat{w}_i - \hat{A}_i - \psi \hat{L}_i^e). \quad (42)$$

Migration channel. *In steady state, population satisfies $L_i = \sum_o m_{oi} n_o L_o$, where $m_{oi} \equiv \frac{B_i(d_{oi}^{-1} V_i)^\eta}{\sum_{k=1}^N B_k(d_{ok}^{-1} V_k)^\eta}$ is the migration probability. Define the expected utility change for agents born in region o as $\tilde{V}_o \equiv \sum_k m_{ok} \hat{V}_k$. Log-linearizing around steady state:*

$$\hat{L}_i = \sum_o \pi_{oi} \left[\eta (\hat{V}_i - \tilde{V}_o) + \hat{n}_o + \hat{L}_o \right], \quad (43)$$

where $\pi_{oi} \equiv m_{oi} n_o L_o / L_i$ is the share of region i 's population originating from region o .

Housing price channel. *Population and per-capita housing demand changes affect rents through the housing supply curve:*

$$\hat{r}_i = \xi_i (\hat{h}_i + \hat{L}_i), \quad (44)$$

where ξ_i is the inverse housing supply elasticity.

Proof. See Appendix A.2. □

These three channels interact to determine the spatial propagation of shocks. In the region receiving a positive productivity shock, wages rise and migrants arrive, bidding up housing costs. Both higher wages and higher rents reduce fertility—wages through the substitution effect, rents through the price effect—generating a fertility decline in booming regions.

In other regions, the effects are more nuanced. Trading partners benefit from lower import prices through equation (41), which raises real income and may increase fertility. Regions losing migrants experience reduced housing demand. Through equation (44), this translates into lower rents, which raises fertility through the price effect. The net effect on aggregate fertility depends on whether these offsetting forces balance.

A key implication is that endogenous fertility acts as a stabilizing force in spatial dynamics.

Proposition 3 (Fertility as a Stabilizing Force). *Consider a positive productivity shock in region j that raises local wages and rents, thereby reducing local fertility: $\hat{n}_j < 0$. Comparing steady states—defined as equilibria where population shares are constant—before and after the shock:*

$$\hat{L}_j^{SS} \big|_{\text{endogenous } n} < \hat{L}_j^{SS} \big|_{\text{exogenous } n} \quad (45)$$

That is, endogenous fertility dampens the long-run population response to productivity shocks.

Proof. See Appendix A.3. □

This result helps explain why highly productive cities have not grown even faster despite sustained migration inflows. The migration-induced increase in housing costs depresses fertility, partially offsetting population growth through natural increase. Endogenous fertility thus provides an equilibrating mechanism that moderates spatial concentration. The magnitude of this dampening depends on the natural retention rate $\mu_j \equiv m_{jj}n_j$ —the fraction of population that stays and reproduces locally—with larger μ_j implying stronger stabilization.

In sum, Proposition 2 and 3 suggest that:

1. *Local shocks propagate across space* through trade (affecting price indices), migration (relocating population), and housing markets (translating population changes into rent changes). These channels generate heterogeneous fertility responses across regions even to localized shocks.

2. *Endogenous fertility stabilizes spatial dynamics* by dampening population responses to productivity shocks. Regions experiencing booms see fertility decline, partially offsetting migration-driven population growth.

3.3 Aggregate Fertility and the Extensive Margin

Aggregate fertility is a population-weighted average of local fertility rates. Changes in aggregate fertility therefore reflect both local fertility changes (the intensive margin) and population reallocation across locations with different fertility rates (the extensive margin).

Proposition 4 (Decomposition of Aggregate Fertility). *Aggregate fertility $\bar{n} \equiv \sum_i n_i L_i / \bar{L}$, where $\bar{L} \equiv \sum_i L_i$, evolves according to:*

$$d\bar{n} = \underbrace{\sum_i \iota_i dn_i}_{\text{intensive margin}} + \underbrace{\sum_i (n_i - \bar{n}) d\iota_i}_{\text{extensive margin}}, \quad (46)$$

where $\iota_i \equiv L_i / \bar{L}$ is region i 's population share. In terms of log changes:

$$\hat{\bar{n}} = \sum_i \frac{\iota_i n_i}{\bar{n}} \hat{n}_i + \frac{1}{\bar{n}} \text{Cov}_s(n_i, \hat{L}_i), \quad (47)$$

where $\text{Cov}_s(n_i, \hat{L}_i) \equiv \sum_i \iota_i (n_i - \bar{n}) \hat{L}_i$ is the population-weighted covariance between fertility levels and population growth rates. The extensive margin is negative when population flows toward regions with below-average fertility.

Proof. See Appendix A.4. □

The decomposition reveals a potential tension between allocative efficiency and aggregate fertility. Policies that reallocate population toward high-productivity, high-wage, expensive locations may raise aggregate output while reducing aggregate fertility through composition effects. The extensive margin $\text{Cov}_s(n_i, \hat{L}_i) / \bar{n}$ is negative precisely when high-growth regions have low fertility—the empirical pattern documented in Figure 1.

Importantly, this tradeoff is not inevitable. Policies reducing the cost of living in productive regions can raise local fertility through the price effect while simultaneously attracting migrants.

When a policy raises fertility in locations that are gaining population—that is, when the intensive margin raises n_i in regions where $\hat{L}_i > 0$ —the two margins reinforce rather than offset each other. This observation motivates our counterfactual analysis of housing supply reform: by reducing rents in supply-constrained productive cities, such reform raises fertility precisely where population growth is concentrated.

In sum, *aggregate fertility depends on both the intensive and extensive margins*. Policies that attract population to low-fertility locations reduce aggregate fertility through composition effects, but policies that reduce costs in productive cities can align the two margins.

3.4 Welfare-Increasing Reallocations

How do spatial transfers $(T_i)_{i=1}^N$ affect expected welfare W and population growth g ? How would a welfare-maximizing social planner choose transfers $(T_i)_{i=1}^N$ and, through them, the spatial population distribution $(L_i)_{i=1}^N$? What are the implications of welfare-maximizing transfers for population growth g ? To obtain analytical insights into these questions, we characterize the equilibrium and optimal policy in the following simplified economy:

Definition 4 (A Simplified Economy). *Suppose that goods are freely traded, i.e. $\tau_{in} = 1$ for all (i, n) , and therefore $P_n = P = 1$ for all n . Suppose that $\xi_i \rightarrow \infty$ for all i , so that the housing supply is inelastic with $H_i = 1$ for all i . Suppose that there is no migration cost, i.e. $d_{oi} = 1$ for all (o, i) . Suppose that $\gamma = 1$ and $\rho \in (0, 1)$. Suppose that $\zeta = 0$, so that χ_i is exogenous. Finally, suppose that the elasticity of substitution across varieties satisfies $\sigma \rightarrow \infty$. In this case, due to no arbitrage, we have $w_i = A_i(L_i^e)^\psi$ for all i .*

This simplified economy preserves the central mechanisms of our model, including the agglomeration externality captured by ψ , the congestion force governed by η , endogenous fertility, and the complementarity between children and housing. At the same time, it abstracts from inter-regional trade networks, bilateral migration frictions, and elastic housing supply, allowing for a tractable analytical characterization of the equilibrium. In the simplified economy in Definition 4, (L_i, g) satisfy

$$\frac{gL_i}{\sum_{o=1}^N n_o L_o} = \frac{B_i V_i^\eta}{\sum_{o=1}^N B_o V_o^\eta}, \quad g = \frac{\sum_{i=1}^N n_i L_i}{\sum_{i=1}^N L_i} \Rightarrow \frac{L_i}{\sum_{o=1}^N L_o} = \frac{B_i V_i^\eta}{\sum_{o=1}^N B_o V_o^\eta}. \quad (48)$$

Given the policy (T_i) and normalizing $\sum_{o=1}^N L_o = 1$, the equilibrium of this simplified economy consists of (L_i, n_i, r_i) such that

$$L_i = \frac{B_i V_i^\eta}{\sum_{o=1}^N B_o V_o^\eta}, \quad (49)$$

where

$$V_i = \frac{w_i + T_i}{\left[\beta r_i^{1-\rho} + (1-\beta)(\chi_i w_i)^{1-\rho} \right]^{\frac{1-\alpha}{1-\rho}}}, \quad (50)$$

and

$$w_i = A_i [(1 - \chi_i n_i) L_i]^\psi. \quad (51)$$

r_i satisfies the housing market clearing:

$$\frac{\beta r_i^{1-\rho}}{\beta r_i^{1-\rho} + (1-\beta)(\chi_i w_i)^{1-\rho}} (1-\alpha)(w_i + T_i) L_i = r_i. \quad (52)$$

n_i satisfies

$$n_i = \frac{(1-\beta)(\chi_i w_i)^{1-\rho}}{\beta r_i^{1-\rho} + (1-\beta)(\chi_i w_i)^{1-\rho}} (1-\alpha)(w_i + T_i). \quad (53)$$

Spatial transfers $(T_i)_{i=1}^N$ are subject to the following budget constraint:

$$\sum_{i=1}^N T_i L_i = \sum_{i=1}^N r_i. \quad (54)$$

The aggregate welfare satisfies:

$$W = \left[\sum_{i=1}^N B_i V_i^\eta \right]^{\frac{1}{\eta}}. \quad (55)$$

Population growth satisfies

$$g = \sum_{i=1}^N n_i L_i. \quad (56)$$

We first consider a benchmark special case in which $\psi = 0$ and $\rho = 1$. In this case, the spatial transfers that maximize welfare W have the following simple form:

Lemma 1 (Benchmark with $\psi = 0$ and $\rho = 1$). *For any $Z > 0$, let $\hat{Z} = dZ/Z$. Consider the simplified economy in Definition 4. Suppose that $\psi = 0$ and $\rho = 1$. Then the spatial transfers $(T_i)_{i=1}^N$ that maximize welfare W satisfy*

$$w_i + (1 + \eta)T_i = D \equiv \frac{1 + \eta\beta(1 - \alpha)}{1 - \beta(1 - \alpha)} \sum_{o=1}^N w_o L_o, \quad \forall i. \quad (57)$$

But, in general, welfare-maximizing transfers $(T_i)_{i=1}^N$ do not maximize population growth g , since $(T_i)_{i=1}^N$ that maximize g satisfy:

$$\frac{1 + \eta}{\sum_{o=1}^N (w_o + T_o) L_o} (w_i + T_i) - \eta [1 - \beta(1 - \alpha)] \propto w_i + (1 + \eta)T_i, \quad \forall i. \quad (58)$$

Lemma 2 (Reallocation Effects). *Consider the simplified economy in Definition 4. Changes in nominal wages can be expressed as*

$$\hat{w}_i = \psi (\hat{L}_i - \delta_i \hat{n}_i), \quad \delta_i \equiv \frac{\chi_i n_i}{1 - \chi_i n_i}. \quad (59)$$

Changes in fertility can be given by

$$\hat{n}_i = \epsilon_i^{n,L} \hat{L}_i + \epsilon_i^{n,T} \hat{T}_i - \epsilon_i^{n,r} \hat{r}_i, \quad (60)$$

where

$$\epsilon_i^{n,L} \equiv \frac{\psi (s_i^w + (1 - \rho) \tilde{\omega}_i^h)}{1 + \psi \delta_i (s_i^w + (1 - \rho) \tilde{\omega}_i^h)}, \quad \epsilon_i^{n,T} \equiv \frac{s_i^T}{1 + \psi \delta_i (s_i^w + (1 - \rho) \tilde{\omega}_i^h)}, \quad \epsilon_i^{n,r} \equiv \frac{(1 - \rho) \tilde{\omega}_i^h}{1 + \psi \delta_i (s_i^w + (1 - \rho) \tilde{\omega}_i^h)}, \quad (61)$$

and

$$s_i^w \equiv \frac{w_i}{w_i + T_i}, \quad s_i^T \equiv \frac{T_i}{w_i + T_i}, \quad \tilde{\omega}_i^h \equiv \frac{\beta r_i^{1-\rho}}{\beta r_i^{1-\rho} + (1 - \beta)(\chi_i w_i)^{1-\rho}}, \quad \tilde{\omega}_i^n \equiv 1 - \tilde{\omega}_i^h. \quad (62)$$

Changes in housing rents can be expressed as

$$\hat{r}_i = \epsilon_i^{r,L} \hat{L}_i + \epsilon_i^{r,T} \hat{T}_i, \quad (63)$$

where

$$\varepsilon_i^{r,L} \equiv \frac{1 + \psi(1 - \delta_i \varepsilon_i^{n,L})(s_i^w - (1 - \rho)\tilde{\omega}_i^n)}{\rho + (1 - \rho)\tilde{\omega}_i^h - \psi \delta_i \varepsilon_i^{n,r}(s_i^w - (1 - \rho)\tilde{\omega}_i^n)}, \quad \varepsilon_i^{r,T} \equiv \frac{s_i^T - \psi \delta_i \varepsilon_i^{n,T}(s_i^w - (1 - \rho)\tilde{\omega}_i^n)}{\rho + (1 - \rho)\tilde{\omega}_i^h - \psi \delta_i \varepsilon_i^{n,r}(s_i^w - (1 - \rho)\tilde{\omega}_i^n)}. \quad (64)$$

Proposition 5 (Welfare-Maximizing Transfers). *Consider the simplified economy in Definition 4. The changes in regional utility can be expressed as*

$$\hat{V}_i = \varepsilon_i^{V,L} \hat{L}_i + \varepsilon_i^{V,T} \hat{T}_i, \quad (65)$$

where

$$\varepsilon_i^{V,L} \equiv \psi(s_i^w - (1 - \alpha)\tilde{\omega}_i^n) \left(1 - \delta_i \varepsilon_i^{n,L} + \delta_i \varepsilon_i^{n,r} \varepsilon_i^{r,L} \right) - (1 - \alpha)\tilde{\omega}_i^h \varepsilon_i^{r,L}, \quad (66)$$

and

$$\varepsilon_i^{V,T} \equiv s_i^T + \psi \delta_i (s_i^w - (1 - \alpha)\tilde{\omega}_i^n) \left(\varepsilon_i^{n,r} \varepsilon_i^{r,T} - \varepsilon_i^{n,T} \right) - (1 - \alpha)\tilde{\omega}_i^h \varepsilon_i^{r,T}. \quad (67)$$

The change in aggregate welfare is thus

$$\hat{W} = \sum_{i=1}^N \Lambda_i \hat{T}_i, \quad (68)$$

where

$$\Lambda_i \equiv \frac{\frac{l_i \varepsilon_i^{V,T}}{1 - \eta \varepsilon_i^{V,L}}}{1 + \eta \sum_{o=1}^N \frac{l_o \varepsilon_o^{V,L}}{1 - \eta \varepsilon_o^{V,L}}}, \quad l_i \equiv \frac{L_i}{\sum_{o=1}^N L_o}. \quad (69)$$

Let $s_i^{TL} \equiv \frac{T_i L_i}{R}$, $s_i^r \equiv \frac{r_i}{R}$, and $R \equiv \sum_{i=1}^N r_i$. Then the transfers $(T_i)_{i=1}^N$ that maximize W can be characterized by the following condition:

$$\Lambda_i \propto s_i^{TL} - s_i^r \varepsilon_i^{r,T} + \left(s_i^{TL} - s_i^r \varepsilon_i^{r,L} \right) \frac{\eta \varepsilon_i^{V,T}}{1 - \eta \varepsilon_i^{V,L}}. \quad (70)$$

Proposition 6 (Population Growth under Reallocations). *Consider the simplified economy in Definition 4. The aggregate population growth can be expressed as*

$$\hat{g} = \sum_{i=1}^N \Omega_i \hat{T}_i, \quad (71)$$

where

$$\begin{aligned} \Omega_i \equiv & s_i^n \left[\left(1 + \varepsilon_i^{n,L} - \varepsilon_i^{n,r} \varepsilon_i^{r,L} \right) \frac{\eta \varepsilon_i^{V,T}}{1 - \eta \varepsilon_i^{V,L}} + \varepsilon_i^{n,T} - \varepsilon_i^{n,r} \varepsilon_i^{r,T} \right] \\ & - \left[\sum_{o=1}^N s_o^n \left(1 + \varepsilon_o^{n,L} - \varepsilon_o^{n,r} \varepsilon_o^{r,L} \right) \frac{\eta}{1 - \eta \varepsilon_o^{V,L}} \right] \Lambda_i, \end{aligned} \quad (72)$$

and

$$s_i^n \equiv \frac{n_i L_i}{g}. \quad (73)$$

Therefore, the transfers $(T_i)_{i=1}^N$ that maximizes g can be characterized by

$$\Omega_i \propto s_i^{TL} - s_i^r \varepsilon_i^{r,T} + \left(s_i^{TL} - s_i^r \varepsilon_i^{r,L} \right) \frac{\eta \varepsilon_i^{V,T}}{1 - \eta \varepsilon_i^{V,L}}. \quad (74)$$

4 Calibration

This section describes our calibration strategy. We calibrate the model to match data on wages, population, fertility, housing rents, and trade and migration flows across 741 U.S. commuting zones, drawn from the 2000 U.S. Census via IPUMS USA (Ruggles et al., 2025). The calibration proceeds in five steps. First, we assign values to parameters taken from the existing literature. Second, we estimate trade and migration costs using bilateral flow data. Third, we construct an IV-based housing–fertility moment that helps discipline the relevant preference elasticity. Fourth, we jointly calibrate the preference parameters using simulated method of moments. Finally, we recover region-specific fundamentals to match the observed spatial distribution exactly.

4.1 Data and Predetermined Parameters

Table 1 summarizes the pre-determined parameters and data sources. The agglomeration elasticity $\psi = 0.1$ lies in the middle of the range of estimates in the urban economics literature, implying that doubling local employment raises productivity by approximately 7 percent (Combes et al., 2012; Rosenthal and Strange, 2004). The migration elasticity $\eta = 1.5$ follows Tombe and Zhu (2019), implying that a 10 percent increase in real income increases the migration flow by approximately

15 percent. The trade elasticity $\sigma - 1 = 4$ is standard in the quantitative trade literature (Head and Mayer, 2014).

The inverse housing supply elasticity ξ_i uses metropolitan-area-level estimates from Saiz (2010), based on geographic constraints and land use regulations. For commuting zones without Saiz estimates, we assign the median supply elasticity. The housing supply shifter $\bar{r}_i$ is recovered to match observed rents and quantities.

Table 1: Data and Predetermined Parameters

Predetermined Parameters	
Parameter	Sources
ψ : agglomeration	= 0.1 from the literature
η : migration elasticity	= 1.5 from Tombe and Zhu (2019)
$\sigma - 1$: trade elasticity	= 4 from the literature
ξ_i : inverse supply elasticity of housing	Saiz (2010)
$\bar{r}_i$: housing supply shifter	housing rent and quantity
Data	
Variable	Data Source
λ_{in} : Trade flows across U.S. regions	Commodity Flow Survey and gravity estimation
L_{oi} : Migration flows across U.S. regions	Census 2000 county-to-county migration flows
n_i : Fertility rate	IPUMS USA, Census 2000
w_i : Wage	IPUMS USA, Census 2000
L_i : Population	IPUMS USA, Census 2000
h_i : Per-capita housing	IPUMS USA, Census 2000
r_i : Housing rent	IPUMS USA, Census 2000

4.2 Trade and Migration Costs

We estimate trade costs τ_{in} and migration costs d_{oi} using gravity regressions on bilateral flow data. Both costs take a power-law form in geographic distance.

Trade Costs. We assume $\tau_{in} = D_{in}^{\delta_\tau}$, where D_{in} is the great-circle distance between regions i and n , computed from geographic coordinates in Chetty, Stepner, Abraham, Lin, Scuderi, Turner, Bergeron, and Cutler (2016). The distance elasticity δ_τ follows Ramondo, Rodríguez-Clare, and

Saborío-Rodríguez (2016), who estimate

$$\log \lambda_{in} = -(1 - \sigma) \delta_\tau \log D_{in} + \text{fe}_i + \text{fe}_n + \tilde{u}_{in}, \quad (75)$$

where λ_{in} is the bilateral trade share from state i to n , and the fixed effects absorb origin productivity and destination demand shifters.

Migration Costs. We assume $d_{oi} = D_{oi}^{\delta_d}$ and estimate:

$$\log L_{oi} = -\eta \delta_d \log D_{oi} + \text{fe}_o + \text{fe}_i + \tilde{\epsilon}_{oi}, \quad (76)$$

where L_{oi} is the number of migrants from o to i . The estimated coefficient is $\widehat{\eta \delta_d} = -2.09$ (standard error 0.01), indicating a strongly negative relationship between distance and migration flows. Given $\eta = 1.5$, this implies $\delta_d = 1.4$. Doubling the distance reduces migration flows by approximately a factor of $2^{1.4} \approx 2.6$.

4.3 The Housing–Fertility Moment

The mechanisms in Section 3 assign a central role to the complementarity between housing and children: the parameter ρ governs how strongly local housing conditions transmit into fertility choices. This subsection constructs an IV-based housing–fertility moment across U.S. commuting zones, using differential exposure to national housing-market shocks generated by local geographic constraints on housing supply. The resulting average marginal effect is included among the moments used to discipline ρ and the other preference parameters in Section 4.4.

Data. We construct a three-wave panel of 741 commuting zones (CZs) from the 1980 and 2000 U.S. Censuses and the 2020 American Community Survey, accessed via IPUMS USA (Ruggles et al., 2025) and aggregated to 1990-consistent CZs following Autor and Dorn (2013) and Autor et al. (2019). The dependent variable is the log total fertility rate ($\ln \text{TFR}_{jt}$), defined as births per woman aged 15–45 at the CZ–year level, constructed as described in Appendix B.1. Housing quantity is measured by NB_{jt} , the average number of bedrooms per owner-occupied dwelling in CZ j and year t . The homeownership rate Own_j is the share of owner-occupied dwellings measured

in the 1980 Census and held fixed across waves; using its predetermined 1980 value rather than contemporaneous tenure mitigates the concern that tenure choices respond to fertility plans. All specifications control for log average wages, population density, and average years of schooling, include CZ and year fixed effects, and are weighted by the number of women aged 15–45. Standard errors are clustered at the state level.

Specification. We estimate

$$\ln \text{TFR}_{jt} = \beta_1 \text{NB}_{jt} + \beta_2 \text{NB}_{jt} \times \text{Own}_j + X'_{jt} \Gamma + \mu_j + \tau_t + \varepsilon_{jt}, \quad (77)$$

so that estimation uses within-CZ changes in housing quantity across the three waves. The marginal effect of an additional bedroom is $\beta_1 + \beta_2 \text{Own}_j$. Thus, β_1 is the fitted marginal effect at a homeownership share of zero, while $\beta_1 + \beta_2$ is the corresponding fitted effect at a homeownership share of one. Motivated by the housing-cost and housing-wealth channels discussed in the literature, the interaction allows the estimated housing–fertility relationship to vary with the CZ’s baseline homeownership share (Dettling and Kearney, 2014; Lovenheim and Mumford, 2013).

Instrumental-Variable Strategy. Housing quantity is endogenous to fertility: households adjust dwellings in anticipation of births, and local shocks move housing markets and fertility simultaneously. We therefore instrument NB_{jt} and its interaction with a shift-share design,

$$Z_{jt} = \text{Elasticity}_j \times \text{hp}_t, \quad Z_{jt} \times \text{Own}_j, \quad (78)$$

where Elasticity_j is the housing supply elasticity of Saiz (2010), determined by predetermined geographic features such as water bodies and steep terrain, and hp_t is the national average house price. When national housing demand rises, supply-elastic CZs absorb the shock through quantities, including more and larger dwellings, while supply-inelastic CZs absorb it through prices. The interaction of this differential exposure with predetermined geography provides the excluded variation in local housing quantity. The identifying assumption is that, conditional on the fixed effects and controls, this differential exposure does not affect local fertility through other channels.

Table 2: Housing Quantity and Fertility: OLS and 2SLS Estimates

Dependent variable:	OLS	First stages		2SLS
	ln TFR (1)	NB (2)	NB $\times$ Own (3)	ln TFR (4)
NB	−1.3975** (0.5731)			−1.2200*** (0.3566)
NB $\times$ Own	2.1612** (0.9181)			2.6187*** (0.6792)
Elasticity $\times$ hp		0.0100** (0.0040)	0.0006 (0.0026)	
Elasticity $\times$ hp $\times$ Own		−0.0122** (0.0058)	0.0013 (0.0038)	
Controls	Yes	Yes	Yes	Yes
CZ fixed effects	Yes	Yes	Yes	Yes
Year fixed effects	Yes	Yes	Yes	Yes
Kleibergen–Paap rk Wald F				3.51
Cragg–Donald Wald F				63.30
Underidentification p -value				0.037
Commuting zones	741	741	741	741
Observations	2,223	2,223	2,223	2,223

Notes: The sample is a panel of 741 commuting zones in 1980, 2000, and 2020. NB is the average number of bedrooms per owner-occupied dwelling; Own is the 1980 homeownership rate, time-invariant. Columns (2) and (3) report first-stage regressions of the two endogenous variables on the excluded instruments. All columns control for log average wages, population density, and average years of schooling, and are weighted by the number of women aged 15–45. Standard errors clustered at the state level in parentheses: * $p < 0.10$, ** $p < 0.05$, *** $p < 0.01$.

Results. Table 2 reports the estimates. The first stages follow the pattern implied by the design: rising national prices increase dwelling size in supply-elastic CZs (column 2), and this relationship is attenuated where homeownership is high (the negative interaction). In the second stage (column 4), the two coefficients are individually significant and imply different marginal effects at low and high homeownership shares. Evaluated at a homeownership share of zero, an increase of 0.1 bedrooms per dwelling, roughly the within-CZ variation observed across waves, reduces fertility by about 11.5 percent. Evaluated at a homeownership share of one, the same increase raises fertility by about 15 percent ($\hat{\beta}_1 + \hat{\beta}_2 = 1.40$, s.e.=0.539). The pattern is consistent with housing-cost pressure and housing wealth operating in opposite directions, as suggested by the quasi-experimental literature on housing wealth and fertility (Dettling and Kearney, 2014; Loven-

heim and Mumford, 2013). The OLS estimates in column (1) show the same sign pattern with smaller magnitudes, which may reflect simultaneity between housing and fertility. Two caveats apply. First, the Kleibergen–Paap F statistic is below conventional thresholds for strong instruments, so we use the estimates as a calibration moment rather than interpret them as precise structural magnitudes. The Anderson–Rubin weak-instrument-robust test ($F(2,49)=5.86$, $p=0.0052$) rejects the joint null that the coefficients on the endogenous variables are zero, providing weak-IV-robust evidence against that joint null.

Second, alternative functional forms produce a similar sign pattern. We adopt the semi-log specification because it preserves the count interpretation of bedrooms and delivers the strongest first stage.

From Estimates to Model Targets. The preference-parameter calibration in Section 4.4 uses this estimate as one of its targets. Evaluated at the reference homeownership share used to construct the calibration target, $\overline{\text{Own}} = 0.66$: the estimated average marginal effect of housing quantity on log fertility is

$$\hat{\beta}_1 + \hat{\beta}_2 \overline{\text{Own}} = -1.2200 + 2.6187 \times 0.66 = 0.51, \quad (79)$$

which serves as the housing–fertility moment in the joint calibration of the preference parameters. Its model counterpart is the regression of log fertility on per-capita housing in model-simulated data, and the parameters are chosen so that the simulated coefficient matches (79) together with the other calibration targets. Because the baseline model abstracts from tenure heterogeneity, the calibration uses the average effect rather than the coefficients evaluated at the endpoints of the homeownership distribution.

4.4 Preference Parameters

The key structural parameters are the elasticity of substitution between consumption and the housing-fertility composite (γ), the elasticity between housing and fertility (ρ), the consumption share (α), and the housing share within the housing-fertility composite (β). We calibrate these using simulated method of moments.

Mapping between Parameters and Moments. Table 3 summarizes the moments used in the joint calibration. The correlation between fertility and wages is particularly informative about γ : when $\gamma < 1$ (complementarity), higher wages reduce demand for both housing and children. The regression of log fertility on per-capita housing helps discipline ρ ; we target the IV-based average marginal effect of housing quantity on fertility estimated in Section 4.3. Average fertility is particularly informative about α , while the aggregate rent-to-income ratio helps discipline β . Because all four parameters affect multiple moments, the mapping is joint rather than one-to-one.

Table 3: Parameters and Moments in Calibration

Parameter	Moment
γ : elasticity between consumption and housing-fertility	$\text{corr}(n_i, w_i)$
ρ : elasticity between housing and fertility	regress $\log n_i$ on h_i
α : goods consumption share	average fertility n_i
β : housing rent share	aggregate housing-to-income ratio

Data Moments. We construct:

$$m^{data}(\gamma, \rho, \alpha, \beta) = \begin{pmatrix} \text{corr}(n_i, w_i) \\ \text{reg } \log n_i \text{ on } h_i \\ \frac{1}{N} \sum_{i=1}^N n_i \\ \frac{\sum_{i=1}^N h_i L_i}{\sum_{i=1}^N w_i L_i} \end{pmatrix} = \begin{pmatrix} -.2555 \\ .5100 \\ .9895 \\ .7988 \end{pmatrix}. \quad (80)$$

Treatment of the Time Cost Parameter. By allowing χ_i to vary freely, we could match observed fertility exactly for any preference parameters, leaving them unidentified. To address this, we impose $\chi_i = 0.15$ for all regions during preference parameter calibration, following the average estimate in the literature. This restriction requires the model to account for cross-sectional fertility variation through endogenous responses to wages and housing costs during this stage of the calibration.

Calibration Algorithm.

Algorithm 1. *We use a Newton-Raphson algorithm with an outer loop updating preference parameters and an inner loop solving for equilibrium:*

- Initial guess for $(\gamma, \rho, \alpha, \beta)$. Then the **inner loop**:

1. Initial guess for (A_i, B_i)
2. Compute $L_i^e = (1 - \chi_i n_i) L_i$ from the data
3. Compute (P_i, Ξ_i, C_i) using Equations (29), (26), and (24)
4. Update A_i by

$$A_i = \left[\frac{w_i^\sigma (L_i^e)^{1-\psi(\sigma-1)}}{\sum_{n=1}^N \tau_{in}^{1-\sigma} P_n^\sigma C_n L_n} \right]^{\frac{1}{\sigma-1}} \quad (81)$$

5. Compute $V_i = \left[\alpha^{\frac{1}{\gamma}} C_i^{\frac{\gamma-1}{\gamma}} + (1-\alpha)^{\frac{1}{\gamma}} \left[\beta^{\frac{1}{\rho}} h_i^{\frac{\rho-1}{\rho}} + (1-\beta)^{\frac{1}{\rho}} n_i^{\frac{\rho-1}{\rho}} \right]^{\frac{\rho}{\rho-1} \frac{\gamma-1}{\gamma}} \right]^{\frac{\gamma}{\gamma-1}}$

6. Update B_i by

$$B_i = \frac{g L_i}{\sum_{o=1}^N \frac{\left(\frac{1}{d_{oi}} V_i \right)^\eta}{\sum_{k=1}^N B_k^{(0)} \left(\frac{1}{d_{ok}} V_k \right)^\eta} n_o L_o} \quad (82)$$

7. Iterate until convergence

- Compute simulated fertility n_i^{sim} using Equation (30)
- Compute simulated per-capita housing h_i^{sim} using Equation (28)
- Compute simulated moments:

$$m^{sim}(\gamma, \rho, \alpha, \beta) = \begin{pmatrix} \text{corr}(n_i^{sim}, w_i) \\ \text{reg log } n_i^{sim} \text{ on } h_i^{sim} \\ \frac{\frac{1}{N} \sum_{i=1}^N n_i^{sim}}{\frac{\sum_{i=1}^N h_i^{sim} L_i}{\sum_{i=1}^N w_i L_i}} \end{pmatrix} \quad (83)$$

- Let $f(\gamma, \rho, \alpha, \beta) \equiv m^{sim} - m^{data}$ and J be its Jacobian. Update:

$$(\gamma^{new}, \rho^{new}, \alpha^{new}, \beta^{new})' = (\gamma, \rho, \alpha, \beta)' - J^{-1} \times f(\gamma, \rho, \alpha, \beta) \quad (84)$$

- Iterate until convergence

Calibration Results. Table 4 reports the calibrated values. The elasticity $\gamma = 0.76$ indicates complementarity between consumption and the housing-fertility composite. The elasticity $\rho = 0.38$ indicates strong complementarity between housing and children and plays an important role in the quantitative analysis. When housing becomes expensive, households cut back on both housing and children rather than substituting toward children. The consumption share is $\alpha = 0.67$, and the housing share within the composite is $\beta = 0.58$.

Table 4 also reports the sensitivity of the simulated moments with respect to the parameters. Moments 1 and 2, $\text{corr}(n_i^{\text{sim}}, w_i)$ and the regression coefficient of $\log n_i$ on h_i , are particularly responsive to the elasticities of substitution (γ, ρ) , while moments 3 and 4, $\frac{1}{N} \sum_{i=1}^N n_i^{\text{sim}}$ and $\frac{\sum_{i=1}^N h_i^{\text{sim}} L_i}{\sum_{i=1}^N w_i L_i}$, are particularly responsive to the share parameters (α, β) . The cross-effects also make clear that the parameters are jointly disciplined by the full set of moments.

Table 4: Calibrated Preference Parameters and Their Partial Effects on Simulated Moments

Parameters:	γ	ρ	α	β
Calibrated values:	.7625	.3768	.6686	.5782
m1: $\text{corr}(n_i^{\text{sim}}, w_i)$	-2.2773	-.6654	.8030	.3385
m2: reg $\log n_i^{\text{sim}}$ on h_i^{sim}	.2734	-1.0849	1.4444	-.4609
m3: $\frac{1}{N} \sum_{i=1}^N n_i^{\text{sim}}$	.0806	.4484	-3.0139	-2.6418
m4: $\frac{\sum_{i=1}^N h_i^{\text{sim}} L_i}{\sum_{i=1}^N w_i L_i}$	-.1178	-.1850	-2.3670	1.1325

Notes: The partial effects of parameters on simulated moments are defined, for example, as $\frac{\partial m_1}{\partial \gamma}$, at the calibrated values of parameters.

4.5 Model Fit

Given the calibrated parameters, we recover productivity A_i , amenities B_i , and time costs χ_i so that the model exactly matches observed wages, population, and fertility. In this inversion, wages primarily discipline productivity, population primarily disciplines amenities, and fertility primarily disciplines time costs.

Algorithm 2. We use the following algorithm to recover (A_i, B_i, χ_i) that exactly match data on (w_i, L_i, n_i) :

1. Initial guess for (A_i, B_i, χ_i)

2. Solve steady-state equilibrium (Definition 3), obtaining (w_i, L_i, n_i)

3. Update:

$$\begin{aligned} A_i &= A_i \left(\frac{w_i^{data}}{w_i} \right)^{\iota_A} \\ B_i &= B_i \left(\frac{L_i^{data} / (\sum_{k=1}^N L_k^{data})}{L_i / (\sum_{k=1}^N L_k)} \right)^{\iota_B} \\ \chi_i &= \chi_i \left(\frac{n_i}{n_i^{data}} \right)^{\iota_\chi} \end{aligned} \quad (85)$$

where $\iota_A, \iota_B, \iota_\chi$ govern adjustment speed

4. Iterate until convergence

Armed with the calibrated χ_i , we run the following regression to obtain $(\bar{\chi}_i, \zeta)$:

$$\log \chi_i = \underbrace{\zeta}_{0.0863(\text{s.e. } 0.007)} \log L_i + \log \bar{\chi}_i. \quad (86)$$

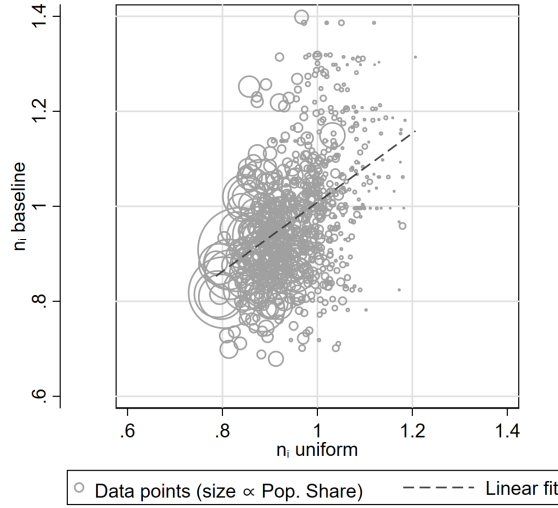
The regression yields $\zeta = 0.083$, suggesting a fertility-specific congestion force in our model.

By construction, the calibrated model exactly matches observed wages, population, and fertility. To assess the model's explanatory power through economic fundamentals alone, Figure 2 plots model-predicted fertility under uniform $\bar{\chi}_i$ against observed fertility. The slope of 0.73 (standard error 0.06) indicates that the model generates meaningful cross-sectional variation even without region-specific fertility shifters. The R^2 of 0.1645 implies that variation generated by the model's economic fundamentals accounts for approximately 16 percent of the observed fertility variance, while the remaining variation is absorbed by the region-specific component of $\bar{\chi}_i$.

We also regress the fertility rate, n_i , simulated under uniform $\bar{\chi}_i$, on simulated wages, population, and housing rents across space. Table 5 shows that, even absent cross-CZ variation in $\bar{\chi}_i$, the model reproduces the negative correlations between fertility and wages, population, and housing rents documented in Figure 1.

Figure 3 compares model-predicted housing rents to observed rents. The slope of 1.0002 (standard error 0.03) indicates no systematic bias, and the R^2 of 0.63 shows that the model explains 61 percent of rent variance. This close fit is important for counterfactual credibility, since housing costs are a key channel affecting fertility.

Figure 2: Model Fit: Fertility under Uniform Parenting Cost Parameter $\bar{\chi}_i$



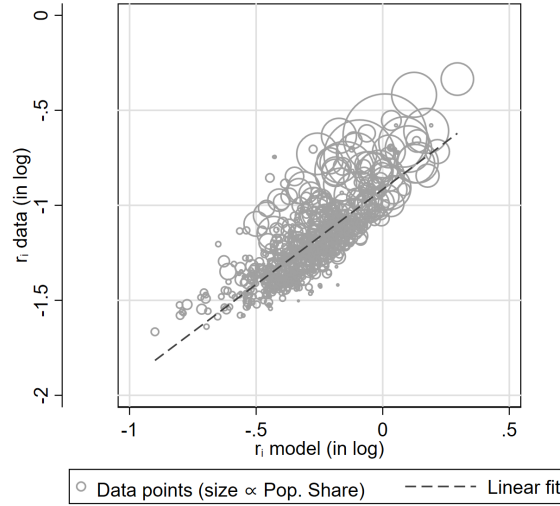
Notes: “ n_i baseline” refers to fertility rates in the baseline model (identical to data). “ n_i uniform” refers to fertility rates under $\bar{\chi}_i = \text{mean}(\bar{\chi}_i)$ for all i . Slope: 0.73 (s.e. 0.06); $R^2 = 0.1645$. Pop. Share refers to population share in the data.

Table 5: Correlations between Fertility, Wage, Population, and Housing Rent under Uniform Parenting Cost Parameter $\bar{\chi}_i$

	Dependent Variable: n_i under uniform $\bar{\chi}_i$		
	(1)	(2)	(3)
$\log(w_i)$ under uniform $\bar{\chi}_i$	-.227*** (.012)		
$\log(L_i)$ under uniform $\bar{\chi}_i$		-.037*** (.001)	
$\log(r_i)$ under uniform $\bar{\chi}_i$			-.369*** (.011)
Observations	741	741	741
R-squared	.279	.686	.630

In summary, the calibrated model matches observed wages, population, and fertility exactly, while delivering a close fit to housing rents. The estimated preference parameters indicate substantial complementarity between housing and fertility, providing a structural foundation for the negative correlation between housing costs and fertility.

Figure 3: Model Fit: Housing Rents



Notes: “ r_i data” refers to observed housing rents. “ r_i model” refers to model-predicted rents. Slope: 1.0002 (s.e. 0.03); $R^2 = 0.63$. Pop. Share refers to population share in the data.

5 Counterfactual Analysis

Armed with the calibrated model, we quantify the effects of regional policies, accounting for endogenous migration and fertility decisions. We first evaluate policies that affect regional housing supply regulations. We then conduct fertility accounting that attributes the changes in fertility rates between 2000 and 2020 to changes in productivities, amenities, and parenting costs. Finally, we characterize interregional transfers that maximize the weighted average of per-capita welfare and aggregate population growth.

5.1 Relaxing Housing Supply Regulations in the Largest 22 CZs

We use the calibrated model to evaluate how housing supply regulations affects fertility, population distribution, and welfare over space. To this end, we relax housing supply regulations in the 22 largest CZs to the median level. In particular, we assign these 22 largest CZs the median value of the Wharton Residential Land Use Regulatory Index (WRLURI), while preserving each location’s observed geographic fundamentals. Applying the structural coefficients from [Saiz \(2010\)](#), we recompute the inverse housing supply elasticities for these 22 largest CZs.

We find that relaxing housing supply regulations in the 22 largest CZs to the median level raises the aggregate steady-state population growth rate, g , from 0.9188 to 0.9281. By reducing housing costs in most productive cities where fertility was previously depressed, the policy raises fertility in locations that account for a large share of the population. Associated with this increase in population, equalizing housing supply regulations to the median increases the expected utility for individuals, W in Equation (34), by 1.83%, mainly due to the increasing population in the most productive CZs with initially tight housing supply regulations.

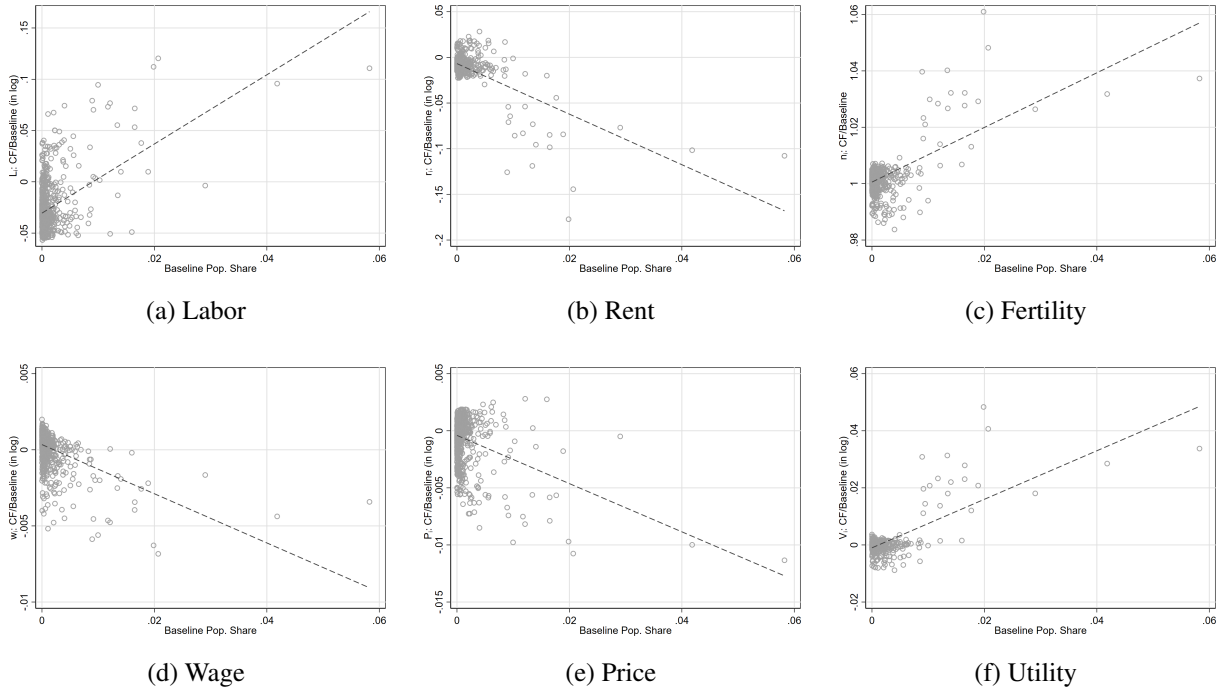
Figure 4 displays the changes in population, housing rent, fertility, wage, price, and utility across CZs, associated with the baseline population distribution. We find, shown in Panel (a), that population increases disproportionately in CZs that were initially large. This reallocation reflects the core mechanism of our model: relaxing housing supply constraints in productive cities initially reduces rents (Panel (b)) and thereby attracts migrants. The largest population gains accrue to CZs in the upper right of the figure—those with high initial population shares—while smaller CZs experience modest population declines as workers relocate toward now-more-affordable productive centers.

Panel (c) of Figure 4 shows the corresponding pattern for fertility. Fertility increases in the large CZs where housing supply constraints have been relaxed. The mechanism operates through the housing-fertility complementarity documented in our calibration: with $\rho = 0.3768$, housing and children are strong complements, so reducing housing costs raises demand for both. In the previously supply-constrained large CZs, the policy-induced rent decline translates directly into higher fertility through Equation (30). Meanwhile, fertility changes are more muted, and in some cases negative, in smaller CZs whose regulation is tightened by the policy.

The positive relationship between baseline population share and fertility gains visible in Panel (c) illustrates why this policy raises aggregate fertility rather than merely redistributing it. Unlike policies that attract households into low-fertility environments—thereby reducing aggregate fertility through compositional effects—relaxing housing supply regulations raises fertility precisely in the locations receiving population inflows.

As workers moving into initially large, productive, and high-wage CZs, these CZs experience significant decline in wages (Panel (d) of Figure 4), resulting in spatial convergence in nominal wage. Due to the agglomeration effect in production, these labor-in-flow CZs also incur significant decline in consumption price indices (Panel (e) of Figure 4). Combining the effects on wage,

Figure 4: Changes in Steady-State Equilibrium Outcomes: Relaxing Housing Supply Regulations in the 22 Largest CZs to the Median of All CZs



Notes: We take the average nominal wage as the numeraire. Baseline pop. share refers to $\frac{L_i}{\sum_{k=1}^N L_k}$ in our baseline model.

“ L_i : CF/Baseline (in log)” refers to $\log(L_i^{CF}/L_i^{Baseline})$. “ n_i : CF/Baseline” refers to $n_i^{CF}/n_i^{Baseline}$. “ w_i : CF/Baseline (in log)” refers to $\log(w_i^{CF}/w_i^{Baseline})$. “ r_i : CF/Baseline (in log)” refers to $\log(r_i^{CF}/r_i^{Baseline})$. “ P_i : CF/Baseline (in log)” refers to $\log(P_i^{CF}/P_i^{Baseline})$. “ V_i : CF/Baseline (in log)” refers to $\log(V_i^{CF}/V_i^{Baseline})$. The aggregate population growth, g , increases from the baseline level of 0.9188 to the counterfactual level of 0.9281, associated with the increase in aggregate welfare by 1.83%.

consumption price, and rent, equalizing housing supply regulations to the median raises the utility, V_i , in the initially large CZs (Panel (f) of Figure 4), transforming into the aggregate welfare gain of 1.83%.

We further decompose the aggregate welfare gain into consumption, housing, and fertility terms:

$$\hat{W} = \underbrace{\sum_{o=1}^N \sum_{i=1}^N s_o^W m_{oi} \omega_i^C \hat{C}_i}_{\text{consumption}} + \underbrace{\sum_{o=1}^N \sum_{i=1}^N s_o^W m_{oi} \omega_i^h \hat{h}_i}_{\text{housing}} + \underbrace{\sum_{o=1}^N \sum_{i=1}^N s_o^W m_{oi} \omega_i^n \hat{n}_i}_{\text{fertility}}, \quad (87)$$

where

$$s_o^W \equiv \frac{\iota_o W_o}{\sum_{k=1}^N \iota_k W_k}, \quad m_{oi} \equiv \frac{B_i \left(\frac{1}{d_{oi}} V_i \right)^\eta}{\sum_{k=1}^N B_k \left(\frac{1}{d_{ok}} V_k \right)^\eta}. \quad (88)$$

The details of this decomposition are presented in Appendix A.6.

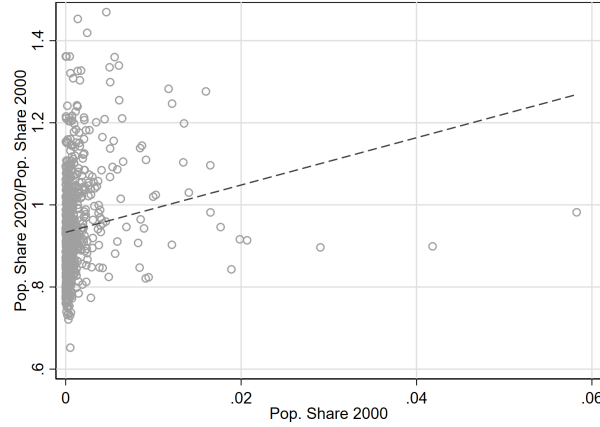
Inserting the shares $(s_o^W, m_{oi}, \omega_i^C, \omega_i^h, \omega_i^n)$ in the baseline and counterfactual changes $(\hat{C}_i, \hat{h}_i, \hat{n}_i)$ into Equation (87), we find that changes in goods consumption contribute to 11.2% of the welfare change, changes in housing contribute to 80.6% of the welfare change, and changes in fertility contribute to 8.2% of the welfare change.

5.2 Fertility Accounting: 2000-2020

In this subsection, we quantify the contributions of changes in productivity A_i , amenities B_i , and parenting costs $\bar{\chi}_i$ to the observed changes in fertility rates between 2000 and 2020. To this end, we calibrate $(A_i, B_i, \bar{\chi}_i)$ to match the 2020 data using the method described in Section 4.5, while holding $(\gamma, \rho, \alpha, \beta, \zeta, \xi_i)$ fixed at their 2000 values. Figure 5 shows that the population became increasingly concentrated in larger CZs between 2000 and 2020, with changes in population shares positively correlated with their initial levels. Given the strong complementarity between housing and fertility in our estimated model, this spatial concentration tends to reduce fertility in large CZs.

We start from the calibrated 2000 economy and separately introduce the changes in (A_i, B_i) and $\bar{\chi}_i$ observed between 2000 and 2020. Panel (a) of Figure 6 shows that changes in productivity and amenities alone generate regional TFRs that are highly correlated with the data, accounting for 36.6% of the variation in changes in TFRs across regions. Panel (b) shows that changes in parenting costs alone explain an even larger share of the variation, accounting for 95.8% of the

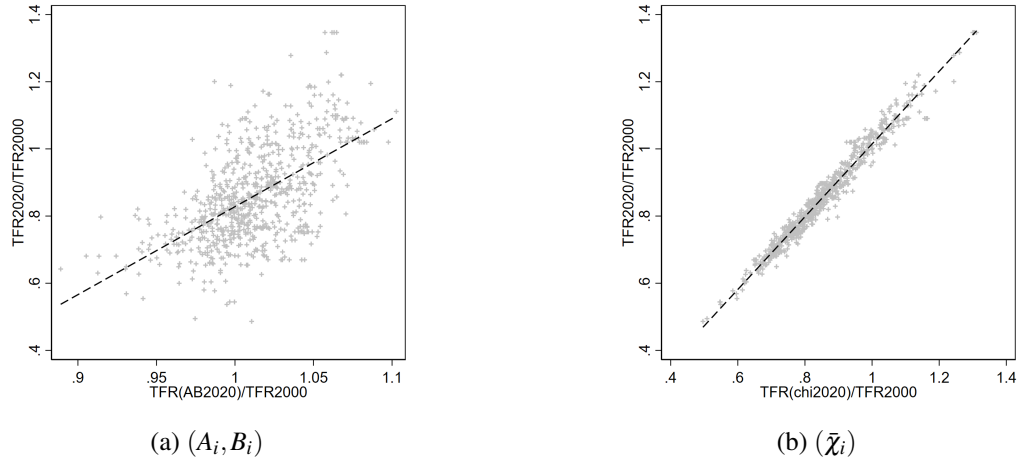
Figure 5: Changes in Population Shares vs. Initial Population Shares



Notes: The slope coefficient is 5.76, with the standard error 2.24.

variation in changes in TFRs across regions. The two channels are not additive: together, their explanatory power exceeds 100%, indicating that they act as substitutes in explaining changes in TFRs across regions between 2000 and 2020.

Figure 6: Fertility Accounting: the Impacts of $(A_i, B_i, \bar{\chi}_i)$ on Fertility Rates across Space



Notes: We start from the economy in 2000. In Panel (a), we replace (A_i, B_i) with their values in 2020, leading to $g = 0.9067$. The linear fit in Panel (a) has the coefficient 2.384 (s.e. 0.110) with $R^2 = 0.366$. In Panel (b), we replace $(\bar{\chi}_i)$ with their values in 2020, leading to $g = 0.7286$. The linear fit in Panel (b) has the coefficient 1.107 (s.e. 0.010) with $R^2 = 0.958$. The actual g in 2020 is 0.7164.

5.3 Optimal Spatial Transfers

In this subsection, we consider spatial transfers $(t_n)_{n=1}^N$ in the steady state:

$$T_n = \frac{t_n}{L_n} \sum_{k=1}^N r_k H_k, \quad \sum_{n=1}^N t_n = 1, \quad t_n \in [0, 1]. \quad (89)$$

Notably, our baseline calibration assumes that $t_n = \frac{L_n}{\sum_{k=1}^N L_k}$.

The social planner is assumed to maximize a linear combination of $\log(W)$, defined by Equation (34), and $\log(g)$, balancing the expected utility for individuals and the population growth rate. The social planner's problem:

$$\begin{aligned} & \max_{\{(t_i; V_i, w_i, r_i, P_i, n_i, L_i)_{i=1}^N, g, W\}} \mathcal{O} \equiv \omega \log W + (1 - \omega) \log(g), \quad \omega \in [0, 1], \\ & \text{s.t.} \\ & (V_i, w_i, r_i, P_i, n_i, L_i, g) \text{ satisfy equilibrium conditions in Definition 3} \\ & W \text{ is defined by Equation (34)} \end{aligned} \quad (90)$$

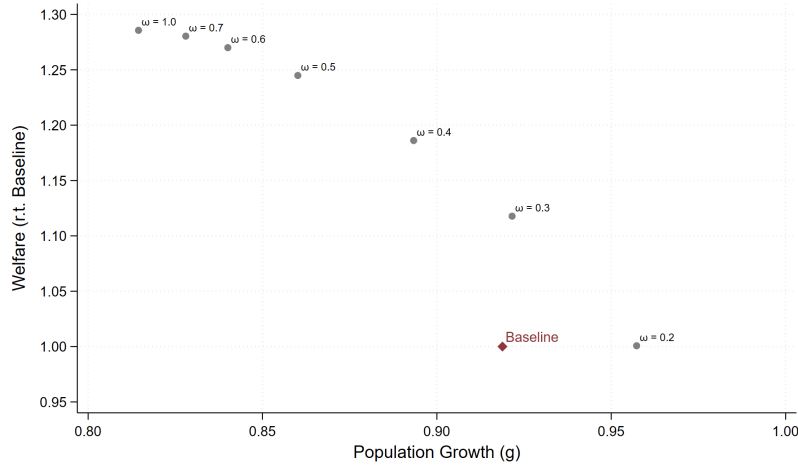
Algorithm 3. *Iterative algorithm to solve for the optimal spatial transfers:*

1. *Initial guess* (t_i)
2. *Compute* $\varepsilon_{\mathcal{O}, t_i} \equiv \frac{\partial \mathcal{O}}{\partial \log t_i}$
3. *Update* $t_i = \frac{\varepsilon_{\mathcal{O}, t_i}}{\sum_{k=1}^N \varepsilon_{\mathcal{O}, t_k}}$
4. *Iterate until convergence*

Figure 7 shows the frontier between per-capita welfare and population growth under optimal interregional transfers with different objectives. Relative to the baseline economy, transfers that maximize per-capita welfare increase welfare by about 30% but reduce the aggregate population growth rate from 0.9188 to about 0.81. In contrast, when $\omega = 0.2$ —that is, when the social planner places a high weight on population growth—transfers increase the population growth rate to about 0.96, while per-capita welfare remains close to its baseline level.

Figure 8 compares the distributions of optimal transfers under $\omega = 1$ and $\omega = 0.2$. When $\omega = 1$, the social planner concentrates transfers in a small number of large cities, exploiting agglomeration

Figure 7: Welfare and Population Growth under the Optimal Transfers



Notes: Baseline refers to the transfer $t_n = \frac{L_n}{\sum_{k=1}^N L_k}$. Welfare W is defined by Equation (34). ω is the share of welfare in the social planner's objective in Problem (90).

economies to raise per-capita real income. When $\omega = 0.2$, transfers are tilted toward small cities, reallocating workers to low-housing-cost locations to increase fertility.

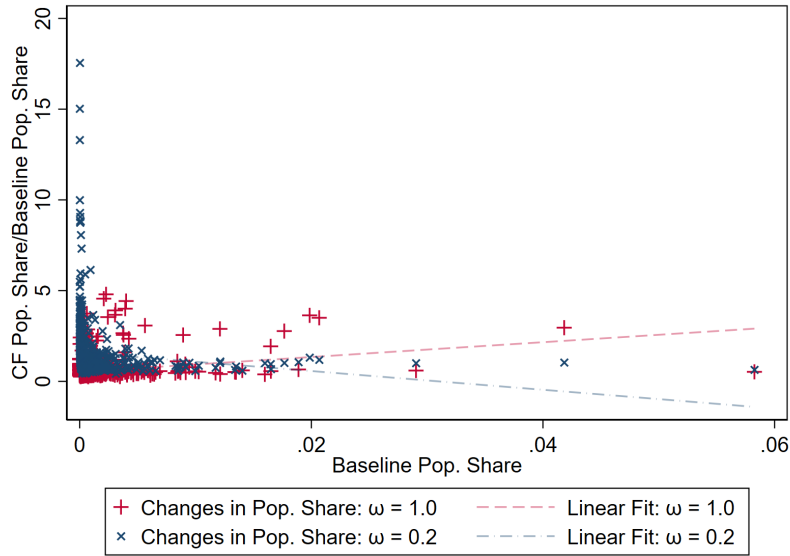


Figure 8: Changes in Population Shares under Optimal Transfers: $\omega = 1.0$ and $\omega = 0.2$

Notes: Baseline refers to the transfer $t_n = \frac{L_n}{\sum_{k=1}^N L_k}$. "CF Pop. share" refers to the counterfactual population share under optimal transfers.

6 Conclusion

This paper develops a quantitative spatial equilibrium model with endogenous fertility to study the interaction between demographic dynamics and the spatial distribution of economic activity. We document that fertility rates are strongly negatively correlated with wages and housing costs across U.S. commuting zones, and we provide a structural interpretation of these patterns through a model in which households jointly choose location, consumption, housing, and fertility.

Our framework integrates household decisions in a multi-region economy with trade, migration, and heterogeneous housing supply. The key insight is that housing and children are complements in household preferences: when housing becomes expensive, households reduce both housing consumption and fertility. We estimate this complementarity to be substantial, with an elasticity of substitution between housing and fertility of 0.38. This parameter implies that housing supply constraints not only reduce welfare through higher living costs but also depress fertility, with consequences for long-run population dynamics.

The model yields several analytical results illuminating the mechanisms at work. Local fertility elasticities decompose into within-nest and cross-nest substitution effects plus income effects, with the relative magnitudes determined by the elasticities of substitution γ and ρ . Cross-place spillovers operate through trade, migration, and price channels, so that shocks in one region affect fertility throughout the economy. Endogenous fertility acts as a stabilizing force in spatial dynamics, accelerating convergence to steady state as regions receiving population inflows experience rent increases that moderate fertility and dampen further growth.

Our counterfactual analysis demonstrates the quantitative importance of housing supply constraints for demographic outcomes. Relaxing housing supply regulations in the 22 largest CZs to the median of all CZs raises the aggregate steady-state population growth from 0.9188 to 0.9281. This effect arises because the policy raises fertility precisely in the large, productive commuting zones that account for substantial population shares. The intensive margin (higher fertility in supply-constrained locations) and extensive margin (population reallocation toward those locations) reinforce rather than offset each other.

These findings carry implications for policy. First, housing supply reform may be a powerful lever for addressing demographic decline. By reducing housing costs in productive cities,

such reforms raise fertility where it matters most for aggregate outcomes. Second, the efficiency-demography trade-off emphasized in prior work is not inevitable. Policies that reduce the cost of living in productive cities can simultaneously improve allocative efficiency and raise aggregate fertility, provided they operate on costs directly rather than simply making cities more attractive along other dimensions. Third, our results complement the findings of [Hsieh and Moretti \(2019\)](#) on housing supply and spatial misallocation: the welfare costs of housing regulation include not only static efficiency losses but also dynamic demographic consequences that compound over time.

We also characterize the optimal interregional transfers. We find that transfers that maximize per-capita welfare concentrate workers in a small number of large CZs, dramatically increasing per-capita real income but reducing fertility rates. In contrast, transfers that place greater weight on population growth retain workers in a large number of small CZs, balancing per-capital real income and population growth.

Several extensions would be valuable. Incorporating heterogeneity in worker skills would allow us to study how the efficiency-demography trade-off varies across the skill distribution and how skill-biased spatial sorting affects aggregate fertility. Adding dynamics of human capital formation would connect our framework to the literature on growth and intergenerational mobility. Extending the model to include international migration would enable analysis of how immigration policy interacts with domestic demographic dynamics. Modeling education competition, as in [Arkolakis et al. \(2026\)](#), would allow us to study an additional channel through which urban density affects the cost of children. We leave these extensions for future work.

References

- ALLEN, T. AND C. ARKOLAKIS (2014): “Trade and the Topography of the Spatial Economy,” *The Quarterly Journal of Economics*, 129, 1085–1140.
- ARKOLAKIS, C., J. H. KWAK, J. WOO, AND H. YANG (2026): “Revisiting the Fertility Puzzle: A Dynamic Spatial Equilibrium Approach,” *Working paper*.
- AUTOR, D., D. DORN, AND G. HANSON (2019): “When work disappears: Manufacturing decline and the falling marriage market value of young men,” *American Economic Review: Insights*, 1, 161–178.
- AUTOR, D. H. AND D. DORN (2013): “The Growth of Low-Skill Service Jobs and the Polarization of the US Labor Market,” *American Economic Review*, 103, 1553–1597.
- BARRO, R. J. AND G. S. BECKER (1989): “Fertility choice in a model of economic growth,” *Econometrica*, 57, 481–501.
- BECKER, G. S. (1960): “An Economic Analysis of Fertility,” in *Demographic and Economic Change in Developed Countries*, Columbia University Press, 209–240.
- BECKER, G. S. AND H. G. LEWIS (1973): “On the Interaction between the Quantity and Quality of Children,” *Journal of Political Economy*, 81, S279–S288.
- BRYAN, G. AND M. MORTEN (2019a): “The Aggregate Productivity Effects of Internal Migration: Evidence from Indonesia,” *Journal of Political Economy*, 127, 2229–2268.
- (2019b): “The Aggregate Productivity Effects of Internal Migration: Evidence from Indonesia,” *Journal of Political Economy*, 127, 2135–2174.
- CALIENDO, L., F. PARRO, E. ROSSI-HANSBERG, AND P.-D. SARTE (2018): “The Impact of Regional and Sectoral Productivity Changes on the US Economy,” *The Review of Economic Studies*, 85, 2042–2096.
- CHETTY, R., M. STEPNER, S. ABRAHAM, S. LIN, B. SCUDERI, N. TURNER, A. BERGERON, AND D. CUTLER (2016): “The Association Between Income and Life Expectancy in the United States, 2001–2014,” *JAMA*, 315, 1750–1766.

- CICCONI, A. AND R. E. HALL (1996): “Productivity and the Density of Economic Activity,” *American Economic Review*, 86, 54–70.
- COMBES, P.-P., G. DURANTON, L. GOBILLON, D. PUGA, AND S. ROUX (2012): “The Productivity Advantages of Large Cities: Distinguishing Agglomeration from Firm Selection,” *Econometrica*, 80, 2543–2594.
- DETTING, L. J. AND M. S. KEARNEY (2014): “House prices and birth rates: The impact of the real estate market on the decision to have a baby,” *Journal of Public Economics*, 110, 82–100.
- DIXIT, A. K. AND J. E. STIGLITZ (1977): “Monopolistic Competition and Optimum Product Diversity,” *American Economic Review*, 67, 297–308.
- DOEPKE, M., A. HANNUSCH, F. KINDERMANN, AND M. TERTILT (2023): “The Economics of Fertility: A New Era,” *Journal of Economic Literature*, 61, 1316–1395.
- EATON, J. AND S. KORTUM (2002): “Technology, Geography, and Trade,” *Econometrica*, 70, 1741–1779.
- FAJGELBAUM, P., E. MORALES, J. C. SUÁREZ SERRATO, AND O. ZIDAR (2019): “State Taxes and Spatial Misallocation,” *The Review of Economic Studies*, 86, 333–376.
- FAJGELBAUM, P. D. AND C. GAUBERT (2020): “Optimal Spatial Policies, Geography, and Sorting,” *The Quarterly Journal of Economics*, 135, 959–1036.
- GOLDIN, C. (2021): *Career and Family: Women’s Century-Long Journey toward Equity*, Princeton University Press.
- GOTO, H. AND K. MINAMIMURA (2019): “Geography and demography: New economic geography with endogenous fertility,” *The Japanese Economic Review*, 70, 537–568.
- HEAD, K. AND T. MAYER (2014): “Gravity Equations: Workhorse, Toolkit, and Cookbook,” *Handbook of International Economics*, 4, 131–195.
- HSIEH, C.-T. AND E. MORETTI (2019): “Housing constraints and spatial misallocation,” *American Economic Journal: Macroeconomics*, 11, 1–39.

- KRUGMAN, P. (1980): “Scale Economies, Product Differentiation, and the Pattern of Trade,” *American Economic Review*, 70, 950–959.
- LOVENHEIM, M. F. AND K. J. MUMFORD (2013): “Do Family Wealth Shocks Affect Fertility Choices? Evidence from the Housing Market,” *Review of Economics and Statistics*, 95, 464–475.
- MCFADDEN, D. (1973): “Conditional Logit Analysis of Qualitative Choice Behavior,” in *Frontiers in Econometrics*, ed. by P. Zarembka, Academic Press, 105–142.
- MORETTI, E. (2013): “Real wage inequality,” *American Economic Journal: Applied Economics*, 5, 65–103.
- RAMONDO, N., A. RODRÍGUEZ-CLARE, AND M. SABORÍO-RODRÍGUEZ (2016): “Trade, domestic frictions, and scale effects,” *American Economic Review*, 106, 3159–3184.
- REDDING, S. J. (2016): “Goods Trade, Factor Mobility and Welfare,” *Journal of International Economics*, 101, 148–167.
- REDDING, S. J. AND E. ROSSI-HANSBERG (2017): “Quantitative Spatial Economics,” *Annual Review of Economics*, 9, 21–58.
- ROSENTHAL, S. S. AND W. C. STRANGE (2004): “Evidence on the Nature and Sources of Agglomeration Economies,” in *Handbook of Regional and Urban Economics*, ed. by J. V. Henderson and J.-F. Thisse, Elsevier, vol. 4, chap. 49, 2119–2171.
- RUGGLES, S., S. FLOOD, M. SOBEK, D. BACKMAN, G. COOPER, J. A. R. DREW, S. RICHARDS, R. RODGERS, J. SCHROEDER, AND K. C. WILLIAMS (2025): *IPUMS USA: Version 16.0 [dataset]*, Minneapolis, MN: IPUMS.
- SAIZ, A. (2010): “The Geographic Determinants of Housing Supply,” *Quarterly Journal of Economics*, 125, 1253–1296.
- SAMUELSON, P. A. (1958): “An Exact Consumption-Loan Model of Interest with or without the Social Contrivance of Money,” *Journal of Political Economy*, 66, 467–482.
- SATO, Y. (2007): “Economic geography, fertility and migration,” *Journal of Urban Economics*, 61, 372–387.

- SATO, Y. AND K. YAMAMOTO (2005): “Population concentration, urbanization, and demographic transition,” *Journal of Urban Economics*, 58, 45–61.
- TOMBE, T. AND X. ZHU (2019): “Trade, Migration, and Productivity: A Quantitative Analysis of China,” *American Economic Review*, 109, 1843–1872.
- ZHOU, A. (2025): “The macroeconomic consequences of family policies,” *Available at SSRN*.

A Theory

A.1 Proof of Proposition 1

We derive the fertility elasticities from the household's optimal fertility choice. The fertility demand function is:

$$n_i = \frac{1}{\chi_i w_i} \cdot \frac{(1-\beta)(\chi_i w_i)^{1-\rho}}{\Psi_i} \cdot \frac{(1-\alpha)\Psi_i^{\frac{1-\gamma}{1-\rho}}}{\Xi_i} \cdot (w_i + T_i), \quad (\text{A.1})$$

where:

$$\Psi_i \equiv \beta r_i^{1-\rho} + (1-\beta)(\chi_i w_i)^{1-\rho}, \quad (\text{A.2})$$

$$\Xi_i \equiv \alpha P_i^{1-\gamma} + (1-\alpha)\Psi_i^{\frac{1-\gamma}{1-\rho}}. \quad (\text{A.3})$$

Step 1: Log-linearization of fertility demand.

Taking logs of (A.1):

$$\begin{aligned} \log n_i &= -\log(\chi_i w_i) + (1-\rho)\log(\chi_i w_i) - \log \Psi_i + \frac{1-\gamma}{1-\rho} \log \Psi_i - \log \Xi_i + \log(w_i + T_i) + \kappa \\ &= -\rho \log(\chi_i w_i) + \frac{\rho-\gamma}{1-\rho} \log \Psi_i - \log \Xi_i + \log(w_i + T_i) + \kappa, \end{aligned} \quad (\text{A.4})$$

where $\kappa = \log[(1-\beta)(1-\alpha)]$ is a constant.

Step 2: Elasticities of Ψ_i and Ξ_i .

From the definition (A.2):

$$\frac{\partial \log \Psi_i}{\partial \log w_i} = \frac{(1-\beta)(1-\rho)(\chi_i w_i)^{1-\rho}}{\Psi_i} = (1-\rho)\tilde{\omega}_i^n, \quad (\text{A.5})$$

$$\frac{\partial \log \Psi_i}{\partial \log r_i} = \frac{\beta(1-\rho)r_i^{1-\rho}}{\Psi_i} = (1-\rho)\tilde{\omega}_i^h, \quad (\text{A.6})$$

$$\frac{\partial \log \Psi_i}{\partial \log \chi_i} = (1-\rho)\tilde{\omega}_i^n. \quad (\text{A.7})$$

From the definition (A.3), using the chain rule:

$$\frac{\partial \log \Xi_i}{\partial \log \Psi_i} = \frac{(1-\alpha)\Psi_i^{\frac{1-\gamma}{1-\rho}}}{\Xi_i} \cdot \frac{1-\gamma}{1-\rho} = (1-\omega_i^C) \cdot \frac{1-\gamma}{1-\rho}, \quad (\text{A.8})$$

where we used $1-\omega_i^C = (1-\alpha)\Psi_i^{\frac{1-\gamma}{1-\rho}}/\Xi_i$. Therefore:

$$\frac{\partial \log \Xi_i}{\partial \log w_i} = (1-\omega_i^C) \cdot \frac{1-\gamma}{1-\rho} \cdot (1-\rho)\tilde{\omega}_i^n = (1-\gamma)(1-\omega_i^C)\tilde{\omega}_i^n, \quad (\text{A.9})$$

$$\frac{\partial \log \Xi_i}{\partial \log r_i} = (1-\gamma)(1-\omega_i^C)\tilde{\omega}_i^h. \quad (\text{A.10})$$

Step 3: Wage elasticity of fertility.

Differentiating (A.4) with respect to $\log w_i$, holding T_i fixed:

$$\begin{aligned} \varepsilon_{n,w} &= -\rho + \frac{\rho-\gamma}{1-\rho} \cdot (1-\rho)\tilde{\omega}_i^n - (1-\gamma)(1-\omega_i^C)\tilde{\omega}_i^n + \frac{w_i}{w_i+T_i} \\ &= \underbrace{-\rho - (\gamma-\rho)\tilde{\omega}_i^n - (1-\gamma)(1-\omega_i^C)\tilde{\omega}_i^n}_{\text{substitution effect}} + \underbrace{\frac{w_i}{w_i+T_i}}_{\text{income effect}}. \end{aligned} \quad (\text{A.11})$$

The substitution effect can be rewritten to clarify its sign. Factoring out $\tilde{\omega}_i^n$ from the last two terms:

$$-\rho + \tilde{\omega}_i^n \left[-(\gamma-\rho) - (1-\gamma)(1-\omega_i^C) \right]. \quad (\text{A.12})$$

If $\rho \leq \gamma < 1$, then the substitution effect is negative.

Step 4: Rent elasticity of fertility.

Differentiating (A.4) with respect to $\log r_i$, holding T_i fixed:

$$\begin{aligned} \varepsilon_{n,r} &= \frac{\rho-\gamma}{1-\rho} \cdot (1-\rho)\tilde{\omega}_i^h - (1-\gamma)(1-\omega_i^C)\tilde{\omega}_i^h \\ &= \tilde{\omega}_i^h \left[-(\gamma-\rho) - (1-\gamma)(1-\omega_i^C) \right]. \end{aligned} \quad (\text{A.13})$$

It is straightforward that $\varepsilon_{n,r} < 0$ if $\rho \leq \gamma < 1$.

Step 5: Time cost elasticity of fertility.

Differentiating (A.4) with respect to $\log \chi_i$:

$$\varepsilon_{n,\chi} = -\rho - (\gamma - \rho) \tilde{\omega}_i^n - (1 - \gamma)(1 - \omega_i^C) \tilde{\omega}_i^n = \varepsilon_{n,w} - \frac{w_i}{w_i + T_i}. \quad (\text{A.14})$$

This relationship holds because χ_i and w_i enter symmetrically in the opportunity cost $\chi_i w_i$, but only w_i affects income. $\square$

A.2 Proof of Proposition 2

Trade channel.

The price index in region i is:

$$P_i = \left[\sum_{k=1}^N \left(\tau_{ik} \cdot \frac{w_k}{A_k (L_k^e)^\psi} \right)^{1-\sigma} \right]^{\frac{1}{1-\sigma}}. \quad (\text{A.15})$$

Define unit cost $c_k \equiv w_k / [A_k (L_k^e)^\psi]$. The expenditure share is:

$$\lambda_{ik} = \frac{(\tau_{ik} c_k)^{1-\sigma}}{P_i^{1-\sigma}}. \quad (\text{A.16})$$

Taking logs: $(1 - \sigma) \log P_i = \log [\sum_k (\tau_{ik} c_k)^{1-\sigma}]$.

Differentiating with respect to $\log A_j$, holding w_k and L_k^e fixed:

$$(1 - \sigma) \frac{\partial \log P_i}{\partial \log A_j} = \frac{(1 - \sigma) (\tau_{ij} c_j)^{1-\sigma}}{P_i^{1-\sigma}} \cdot \frac{\partial \log c_j}{\partial \log A_j} = (1 - \sigma) \lambda_{ij} \cdot (-1). \quad (\text{A.17})$$

Thus $\partial \log P_i / \partial \log A_j|_{w, L^e} = -\lambda_{ij}$.

In general equilibrium, log-linearizing:

$$\hat{P}_i = \sum_k \lambda_{ik} \hat{c}_k = \sum_k \lambda_{ik} (\hat{w}_k - \hat{A}_k - \psi \hat{L}_k^e). \quad (\text{A.18})$$

Migration channel.

In steady state: $L_i = \sum_o m_{oi} n_o L_o$, where $m_{oi} = B_i(d_{oi}^{-1} V_i)^\eta / \Phi_o$.

Log-linearizing the migration probability:

$$\hat{m}_{oi} = \eta \hat{V}_i - \hat{\Phi}_o = \eta \hat{V}_i - \eta \sum_k m_{ok} \hat{V}_k = \eta (\hat{V}_i - \bar{\hat{V}}_o). \quad (\text{A.19})$$

Log-linearizing population:

$$\hat{L}_i = \sum_o \frac{m_{oi} n_o L_o}{L_i} (\hat{m}_{oi} + \hat{n}_o + \hat{L}_o) = \sum_o \pi_{oi} \left[\eta (\hat{V}_i - \bar{\hat{V}}_o) + \hat{n}_o + \hat{L}_o \right]. \quad (\text{A.20})$$

Housing price channel.

From $r_i = \bar{r}_i H_i^{\xi_i}$: $\log r_i = \log \bar{r}_i + \xi_i \log H_i$, so $\hat{r}_i = \xi_i \hat{H}_i$.

From market clearing $H_i = h_i L_i$: $\hat{H}_i = \hat{h}_i + \hat{L}_i$.

Combining: $\hat{r}_i = \xi_i (\hat{h}_i + \hat{L}_i)$. □

A.3 Proof of Proposition 3

The steady-state population in region j satisfies:

$$L_j = m_{jj} n_j L_j + \sum_{o \neq j} m_{oj} n_o L_o. \quad (\text{A.21})$$

Define the natural retention rate $\mu_j \equiv m_{jj} n_j$ and net immigration $I_j \equiv \sum_{o \neq j} m_{oj} n_o L_o$. Then:

$$L_j = \frac{I_j}{1 - \mu_j}. \quad (\text{A.22})$$

Log-linearizing:

$$\hat{L}_j = \hat{I}_j + \frac{\mu_j}{1 - \mu_j} \hat{\mu}_j = \hat{I}_j + \frac{\mu_j}{1 - \mu_j} (\hat{m}_{jj} + \hat{n}_j). \quad (\text{A.23})$$

With exogenous fertility ($\hat{n}_j = 0$):

$$\hat{L}_j^{SS} \big|_{\text{exo } n} = \hat{I}_j + \frac{\mu_j}{1 - \mu_j} \hat{m}_{jj}. \quad (\text{A.24})$$

With endogenous fertility:

$$\hat{L}_j^{SS}|_{\text{endo } n} = \hat{I}_j + \frac{\mu_j}{1 - \mu_j}(\hat{m}_{jj} + \hat{n}_j). \quad (\text{A.25})$$

The difference is:

$$\hat{L}_j^{SS}|_{\text{endo } n} - \hat{L}_j^{SS}|_{\text{exo } n} = \frac{\mu_j}{1 - \mu_j}\hat{n}_j < 0, \quad (\text{A.26})$$

since $\hat{n}_j < 0$ (productivity shocks raise wages and rents, reducing fertility) and $\mu_j/(1 - \mu_j) > 0$. $\square$

A.4 Proof of Proposition 4

Aggregate fertility is $\bar{n} = \sum_i n_i s_i$ where $s_i = L_i/\bar{L}$ and $\sum_i s_i = 1$.

Taking the total differential:

$$d\bar{n} = \sum_i s_i dn_i + \sum_i n_i ds_i. \quad (\text{A.27})$$

Since $\sum_i ds_i = 0$:

$$\sum_i n_i ds_i = \sum_i n_i ds_i - \bar{n} \sum_i ds_i = \sum_i (n_i - \bar{n}) ds_i. \quad (\text{A.28})$$

For log changes, $ds_i = s_i(\hat{L}_i - \hat{\bar{L}})$ where $\hat{\bar{L}} = \sum_k s_k \hat{L}_k$. The extensive margin:

$$\sum_i (n_i - \bar{n}) s_i (\hat{L}_i - \hat{\bar{L}}) = \sum_i (n_i - \bar{n}) s_i \hat{L}_i = \text{Cov}_s(n_i, \hat{L}_i), \quad (\text{A.29})$$

using $\sum_i (n_i - \bar{n}) s_i = 0$.

Dividing by $\bar{n}$:

$$\hat{n} = \sum_i \frac{s_i n_i}{\bar{n}} \hat{n}_i + \frac{1}{\bar{n}} \text{Cov}_s(n_i, \hat{L}_i). \quad (\text{A.30})$$

$\square$

A.5 Welfare-Increasing Spatial Transfers

A.5.1 Proof to Lemma 1

Proof. Under $\psi = 0$ and $\rho = 1$, we have $w_i = A_i$ and therefore

$$V_i = \frac{A_i + T_i}{r_i^{\beta(1-\alpha)} (\chi_i A_i)^{(1-\beta)(1-\alpha)}}. \quad (\text{A.31})$$

Since $r_i = \beta(1-\alpha)(A_i + T_i)L_i$, we have

$$V_i = \frac{(A_i + T_i)^{1-\beta(1-\alpha)}}{[\beta(1-\alpha)L_i]^{\beta(1-\alpha)} (\chi_i A_i)^{(1-\beta)(1-\alpha)}}. \quad (\text{A.32})$$

Log-linearization leads to

$$\hat{V}_i = [1 - \beta(1 - \alpha)] s_i^T \hat{T}_i - \beta(1 - \alpha) \hat{L}_i. \quad (\text{A.33})$$

Notice that $\hat{L}_i = \eta (\hat{V}_i - \hat{W})$ and $\hat{W} = \sum_{i=1}^N \iota_i \hat{V}_i$ where $\iota_i \equiv L_i / (\sum_{o=1}^N L_o)$. Therefore,

$$\hat{W} = [1 - \beta(1 - \alpha)] \sum_{i=1}^N \iota_i s_i^T \hat{T}_i. \quad (\text{A.34})$$

The budget constraint for transfers can be expressed as

$$\sum_{i=1}^N T_i L_i = \beta(1 - \alpha) \sum_{i=1}^N (A_i + T_i) L_i = R. \quad (\text{A.35})$$

Log-linearizing this condition gives

$$\sum_{i=1}^N s_i^{TL} (\hat{T}_i + \hat{L}_i) = \sum_{i=1}^N s_i^r (s_i^T \hat{T}_i + \hat{L}_i), \quad (\text{A.36})$$

where

$$s_i^{TL} \equiv \frac{T_i L_i}{R}, \quad s_i^r \equiv \frac{\beta(1 - \alpha)(A_i + T_i) L_i}{R}, \quad s_i^T \equiv \frac{T_i}{A_i + T_i}. \quad (\text{A.37})$$

Rearranging yields

$$\sum_{i=1}^N [(s_i^{TL} - s_i^r s_i^T) \hat{T}_i + (s_i^{TL} - s_i^r) \hat{L}_i] = 0. \quad (\text{A.38})$$

Using $\hat{L}_i = \eta(\hat{V}_i - \hat{W})$ and $\hat{V}_i = [1 - \beta(1 - \alpha)] s_i^T \hat{T}_i - \beta(1 - \alpha) \hat{L}_i$, we obtain

$$\hat{V}_i - \hat{W} = \frac{[1 - \beta(1 - \alpha)] s_i^T \hat{T}_i - \hat{W}}{1 + \eta\beta(1 - \alpha)}. \quad (\text{A.39})$$

Therefore,

$$\hat{L}_i = \frac{\eta}{1 + \eta\beta(1 - \alpha)} \{ [1 - \beta(1 - \alpha)] s_i^T \hat{T}_i - \hat{W} \}. \quad (\text{A.40})$$

Substituting into the resource constraint gives

$$\sum_{i=1}^N (s_i^{TL} - s_i^r s_i^T) \hat{T}_i + \frac{\eta}{1 + \eta\beta(1 - \alpha)} \sum_{i=1}^N (s_i^{TL} - s_i^r) \{ [1 - \beta(1 - \alpha)] s_i^T \hat{T}_i - \hat{W} \} = 0. \quad (\text{A.41})$$

Since $\sum_{i=1}^N (s_i^{TL} - s_i^r) = 0$, the term involving $\hat{W}$ disappears. Hence,

$$\sum_{i=1}^N \left[s_i^{TL} - s_i^r s_i^T + (s_i^{TL} - s_i^r) \frac{\eta[1 - \beta(1 - \alpha)] s_i^T}{1 + \eta\beta(1 - \alpha)} \right] \hat{T}_i = 0. \quad (\text{A.42})$$

Hence, the marginal resource cost in the optimality condition is

$$s_i^{TL} - s_i^r s_i^T + (s_i^{TL} - s_i^r) \frac{\eta[1 - \beta(1 - \alpha)] s_i^T}{1 + \eta\beta(1 - \alpha)}, \quad (\text{A.43})$$

which can be expressed as

$$\frac{L_i T_i [1 - \beta(1 - \alpha)]}{R} + \frac{L_i T_i [1 - \beta(1 - \alpha)]}{R} \frac{\eta}{1 + \eta\beta(1 - \alpha)} \left(\frac{T_i}{w_i + T_i} - \beta(1 - \alpha) \right). \quad (\text{A.44})$$

Rearranging,

$$\frac{L_i T_i [1 - \beta(1 - \alpha)]}{R [1 + \eta\beta(1 - \alpha)]} \left(1 + \eta \frac{T_i}{w_i + T_i} \right). \quad (\text{A.45})$$

Therefore, the optimality condition implies

$$L_i \frac{T_i}{w_i + T_i} \propto L_i T_i \left(1 + \eta \frac{T_i}{w_i + T_i} \right). \quad (\text{A.46})$$

Canceling $L_i T_i$ gives

$$\frac{1}{w_i + T_i} \propto 1 + \eta \frac{T_i}{w_i + T_i}. \quad (\text{A.47})$$

Multiplying both sides by $w_i + T_i$ yields

$$w_i + (1 + \eta)T_i = D, \quad (\text{A.48})$$

where D is common across regions.

It remains to determine D . The budget constraint implies

$$\sum_i T_i L_i = \sum_i r_i. \quad (\text{A.49})$$

Using $T_i = (D - w_i)/(1 + \eta)$ and the housing market clearing condition, we obtain

$$\frac{D - \sum_i w_i L_i}{1 + \eta} = \beta(1 - \alpha) \left(\sum_i w_i L_i + \frac{D - \sum_i w_i L_i}{1 + \eta} \right). \quad (\text{A.50})$$

Rearranging gives

$$D = \frac{1 + \eta \beta(1 - \alpha)}{1 - \beta(1 - \alpha)} \sum_{i=1}^N w_i L_i. \quad (\text{A.51})$$

Substituting this expression into $w_i + (1 + \eta)T_i = D$ gives the desired result for welfare-maximizing transfers.

For population-growth-maximizing transfers, notice that

$$\hat{n}_i = s_i^T \hat{T}_i, \quad s_i^T \equiv \frac{T_i}{w_i + T_i}. \quad (\text{A.52})$$

By the definition of population growth,

$$g = \sum_{i=1}^N n_i L_i, \quad (\text{A.53})$$

its log-linearization gives

$$\hat{g} = \sum_{i=1}^N s_i^n (\hat{n}_i + \hat{L}_i), \quad (\text{A.54})$$

where

$$s_i^n \equiv \frac{n_i L_i}{g}. \quad (\text{A.55})$$

Using the equilibrium population response,

$$\hat{L}_i = \frac{\eta}{1 + \eta\beta(1 - \alpha)} ([1 - \beta(1 - \alpha)] s_i^T \hat{T}_i - \hat{W}), \quad (\text{A.56})$$

we obtain

$$\hat{g} = \sum_{i=1}^N s_i^n s_i^T \hat{T}_i + \frac{\eta}{1 + \eta\beta(1 - \alpha)} \sum_{i=1}^N s_i^n ([1 - \beta(1 - \alpha)] s_i^T \hat{T}_i - \hat{W}). \quad (\text{A.57})$$

Since

$$\sum_{i=1}^N s_i^n = 1, \quad (\text{A.58})$$

this becomes

$$\hat{g} = \frac{1 + \eta}{1 + \eta\beta(1 - \alpha)} \sum_{i=1}^N s_i^n s_i^T \hat{T}_i - \frac{\eta}{1 + \eta\beta(1 - \alpha)} \hat{W}. \quad (\text{A.59})$$

From the welfare response derived above,

$$\hat{W} = [1 - \beta(1 - \alpha)] \sum_{i=1}^N L_i s_i^T \hat{T}_i. \quad (\text{A.60})$$

Substituting this expression into $\hat{g}$ gives

$$\hat{g} = \frac{1}{1 + \eta\beta(1 - \alpha)} \sum_{i=1}^N [(1 + \eta)s_i^n - \eta(1 - \beta(1 - \alpha))L_i] s_i^T \hat{T}_i. \quad (\text{A.61})$$

Hence, the marginal effect of increasing T_i on population growth, holding the aggregate transfer constraint into account, is proportional to

$$[(1 + \eta)s_i^n - \eta(1 - \beta(1 - \alpha))L_i] s_i^T. \quad (\text{A.62})$$

Using

$$s_i^n = \frac{(w_i + T_i)L_i}{\sum_{o=1}^N (w_o + T_o)L_o}, \quad (\text{A.63})$$

and

$$s_i^T = \frac{T_i}{w_i + T_i}, \quad (\text{A.64})$$

the marginal growth effect becomes

$$L_i T_i \left[\frac{1 + \eta}{\sum_{o=1}^N (w_o + T_o) L_o} - \frac{\eta(1 - \beta(1 - \alpha))}{w_i + T_i} \right]. \quad (\text{A.65})$$

The marginal resource cost implied by the transfer budget constraint is proportional to

$$L_i T_i \left(1 + \eta \frac{T_i}{w_i + T_i} \right). \quad (\text{A.66})$$

At the growth-maximizing allocation, the marginal growth benefit must be proportional to the marginal resource cost. Therefore,

$$\left[\frac{1 + \eta}{\sum_{o=1}^N (w_o + T_o) L_o} - \frac{\eta(1 - \beta(1 - \alpha))}{w_i + T_i} \right] \propto 1 + \eta \frac{T_i}{w_i + T_i}. \quad (\text{A.67})$$

Multiplying both sides by $w_i + T_i$ yields

$$\frac{1 + \eta}{\sum_{o=1}^N (w_o + T_o) L_o} (w_i + T_i) - \eta(1 - \beta(1 - \alpha)) \propto w_i + (1 + \eta) T_i, \quad (\text{A.68})$$

which is the desired result for population-growth-maximizing transfers. $\square$

A.5.2 Proof to Lemma 2

Proof. For any variable $Z > 0$, define $\hat{Z} = dZ/Z$. Since $w_i = A_i[(1 - \chi_i n_i) L_i]^\psi$, log-linearization immediately gives $\hat{w}_i = \psi(\hat{L}_i - \delta_i \hat{n}_i)$, where $\delta_i \equiv \frac{\chi_i n_i}{1 - \chi_i n_i}$.

Next, consider $n_i = \tilde{\omega}_i^n (1 - \alpha)(w_i + T_i)$, where $\tilde{\omega}_i^n = \frac{(1 - \beta)(\chi_i w_i)^{1 - \rho}}{\beta r_i^{1 - \rho} + (1 - \beta)(\chi_i w_i)^{1 - \rho}}$. Log-linearizing yields $\hat{n}_i = \hat{\tilde{\omega}}_i^n + s_i^w \hat{w}_i + s_i^T \hat{T}_i$, where $s_i^w \equiv \frac{w_i}{w_i + T_i}$ and $s_i^T \equiv \frac{T_i}{w_i + T_i}$. Since $\hat{\tilde{\omega}}_i^n = (1 - \rho) \tilde{\omega}_i^h (\hat{w}_i - \hat{r}_i)$, we obtain $\hat{n}_i = [s_i^w + (1 - \rho) \tilde{\omega}_i^h] \hat{w}_i + s_i^T \hat{T}_i - (1 - \rho) \tilde{\omega}_i^h \hat{r}_i$. Substituting $\hat{w}_i = \psi(\hat{L}_i - \delta_i \hat{n}_i)$ and rearranging gives $\hat{n}_i = \varepsilon_i^{n,L} \hat{L}_i + \varepsilon_i^{n,T} \hat{T}_i - \varepsilon_i^{n,r} \hat{r}_i$, where $\varepsilon_i^{n,L}$, $\varepsilon_i^{n,T}$, and $\varepsilon_i^{n,r}$ are given in the lemma.

Finally, the housing market clearing condition is $r_i = \tilde{\omega}_i^h (1 - \alpha)(w_i + T_i) L_i$. Log-linearizing gives $\hat{r}_i = \hat{\tilde{\omega}}_i^h + s_i^w \hat{w}_i + s_i^T \hat{T}_i + \hat{L}_i$. Since $\hat{\tilde{\omega}}_i^h = (1 - \rho) \tilde{\omega}_i^n (\hat{r}_i - \hat{w}_i)$, we obtain $[\rho + (1 - \rho) \tilde{\omega}_i^h] \hat{r}_i =$

$[s_i^w - (1 - \rho)\tilde{\omega}_i^n]\hat{w}_i + s_i^T \hat{T}_i + \hat{L}_i$. Substituting $\hat{w}_i = \psi(\hat{L}_i - \delta_i \hat{n}_i)$ and $\hat{n}_i = \varepsilon_i^{n,L} \hat{L}_i + \varepsilon_i^{n,T} \hat{T}_i - \varepsilon_i^{n,r} \hat{r}_i$, rearranging terms, and solving for $\hat{r}_i$ gives $\hat{r}_i = \varepsilon_i^{r,L} \hat{L}_i + \varepsilon_i^{r,T} \hat{T}_i$, where $\varepsilon_i^{r,L}$ and $\varepsilon_i^{r,T}$ are given in the lemma. This completes the proof. $\square$

A.5.3 Proof to Proposition 5

Proof. From the definition of regional utility, $V_i = \frac{w_i + T_i}{[\beta r_i^{1-\rho} + (1-\beta)(\chi_i w_i)^{1-\rho}]^{\frac{1-\alpha}{1-\rho}}}$, log-linearization gives $\hat{V}_i = (s_i^w - (1 - \alpha)\tilde{\omega}_i^n)\hat{w}_i + s_i^T \hat{T}_i - (1 - \alpha)\tilde{\omega}_i^h \hat{r}_i$. Substituting $\hat{w}_i = \psi(\hat{L}_i - \delta_i \hat{n}_i)$, $\hat{n}_i = \varepsilon_i^{n,L} \hat{L}_i + \varepsilon_i^{n,T} \hat{T}_i - \varepsilon_i^{n,r} \hat{r}_i$, and $\hat{r}_i = \varepsilon_i^{r,L} \hat{L}_i + \varepsilon_i^{r,T} \hat{T}_i$ from Lemma 2, we obtain $\hat{V}_i = \varepsilon_i^{V,L} \hat{L}_i + \varepsilon_i^{V,T} \hat{T}_i$, where $\varepsilon_i^{V,L}$ and $\varepsilon_i^{V,T}$ are given in the proposition.

The equilibrium allocation condition $L_i = \frac{B_i V_i^\eta}{\sum_o B_o V_o^\eta}$ implies $\hat{L}_i = \eta \hat{V}_i - \eta \sum_o \iota_o \hat{V}_o$, where $\iota_i \equiv L_i / \sum_o L_o$. Substituting $\hat{V}_i = \varepsilon_i^{V,L} \hat{L}_i + \varepsilon_i^{V,T} \hat{T}_i$ gives $\hat{L}_i = \frac{\eta \varepsilon_i^{V,T}}{1 - \eta \varepsilon_i^{V,L}} \hat{T}_i - \frac{\eta}{1 - \eta \varepsilon_i^{V,L}} \sum_o \iota_o \hat{V}_o$. Aggregating across regions and using $\hat{W} = \sum_i \iota_i \hat{V}_i$ yields $\hat{W} = \sum_i \Lambda_i \hat{T}_i$, where Λ_i is defined in the proposition.

Finally, the budget constraint $\sum_i T_i L_i = \sum_i r_i$ implies $\sum_i s_i^{TL}(\hat{T}_i + \hat{L}_i) = \sum_i s_i^r \hat{r}_i$, where $s_i^{TL} \equiv T_i L_i / R$, $s_i^r \equiv r_i / R$, and $R \equiv \sum_i r_i$. Using $\hat{r}_i = \varepsilon_i^{r,L} \hat{L}_i + \varepsilon_i^{r,T} \hat{T}_i$ and the equilibrium response of $\hat{L}_i$ above, the transformed budget constraint becomes

$$\sum_i \left[s_i^{TL} - s_i^r \varepsilon_i^{r,T} + \left(s_i^{TL} - s_i^r \varepsilon_i^{r,L} \right) \frac{\eta \varepsilon_i^{V,T}}{1 - \eta \varepsilon_i^{V,L}} \right] \hat{T}_i = \left[\sum_i \frac{\eta (s_i^{TL} - s_i^r \varepsilon_i^{r,L})}{1 - \eta \varepsilon_i^{V,L}} \right] \hat{W}. \quad (\text{A.69})$$

The term multiplying $\hat{W}$ on the right-hand side is common across regions. Therefore, the marginal resource cost of changing T_i is proportional to $s_i^{TL} - s_i^r \varepsilon_i^{r,T} + (s_i^{TL} - s_i^r \varepsilon_i^{r,L}) \frac{\eta \varepsilon_i^{V,T}}{1 - \eta \varepsilon_i^{V,L}}$. Since the objective function is $\hat{W} = \sum_i \Lambda_i \hat{T}_i$, the first-order condition for maximizing aggregate welfare subject to the transformed budget constraint requires the marginal welfare gain to be proportional to the marginal resource cost. Hence,

$$\Lambda_i \propto s_i^{TL} - s_i^r \varepsilon_i^{r,T} + \left(s_i^{TL} - s_i^r \varepsilon_i^{r,L} \right) \frac{\eta \varepsilon_i^{V,T}}{1 - \eta \varepsilon_i^{V,L}}. \quad (\text{A.70})$$

This completes the proof. $\square$

A.5.4 Proof to Proposition 6

Proof. Since $g = \sum_{i=1}^N n_i L_i$, its log-linearization gives

$$\hat{g} = \sum_{i=1}^N s_i^n (\hat{n}_i + \hat{L}_i), \quad (\text{A.71})$$

where $s_i^n \equiv n_i L_i / g$. Using Lemma 2, we have

$$\hat{n}_i + \hat{L}_i = \left(1 + \varepsilon_i^{n,L} - \varepsilon_i^{n,r} \varepsilon_i^{r,L}\right) \hat{L}_i + \left(\varepsilon_i^{n,T} - \varepsilon_i^{n,r} \varepsilon_i^{r,T}\right) \hat{T}_i. \quad (\text{A.72})$$

From the equilibrium allocation condition,

$$\hat{L}_i = \frac{\eta \varepsilon_i^{V,T}}{1 - \eta \varepsilon_i^{V,L}} \hat{T}_i - \frac{\eta}{1 - \eta \varepsilon_i^{V,L}} \hat{W}. \quad (\text{A.73})$$

Substituting this expression into $\hat{g}$ and using $\hat{W} = \sum_i \Lambda_i \hat{T}_i$ yields

$$\hat{g} = \sum_{i=1}^N \Omega_i \hat{T}_i, \quad (\text{A.74})$$

where

$$\Omega_i \equiv s_i^n \left[\left(1 + \varepsilon_i^{n,L} - \varepsilon_i^{n,r} \varepsilon_i^{r,L}\right) \frac{\eta \varepsilon_i^{V,T}}{1 - \eta \varepsilon_i^{V,L}} + \varepsilon_i^{n,T} - \varepsilon_i^{n,r} \varepsilon_i^{r,T} \right] - \left[\sum_{o=1}^N s_o^n (1 + \varepsilon_o^{n,L} - \varepsilon_o^{n,r} \varepsilon_o^{r,L}) \frac{\eta}{1 - \eta \varepsilon_o^{V,L}} \right] \Lambda_i. \quad (\text{A.75})$$

Finally, the budget constraint implies

$$\sum_i s_i^{TL} (\hat{T}_i + \hat{L}_i) = \sum_i s_i^r \hat{r}_i. \quad (\text{A.76})$$

Using $\hat{r}_i = \varepsilon_i^{r,L} \hat{L}_i + \varepsilon_i^{r,T} \hat{T}_i$ and the equilibrium response of $\hat{L}_i$, the transformed budget constraint becomes

$$\sum_i \left[s_i^{TL} - s_i^r \varepsilon_i^{r,T} + \left(s_i^{TL} - s_i^r \varepsilon_i^{r,L} \right) \frac{\eta \varepsilon_i^{V,T}}{1 - \eta \varepsilon_i^{V,L}} \right] \hat{T}_i = \left[\sum_i \frac{\eta (s_i^{TL} - s_i^r \varepsilon_i^{r,L})}{1 - \eta \varepsilon_i^{V,L}} \right] \hat{W}. \quad (\text{A.77})$$

The term multiplying $\hat{W}$ on the right-hand side is common across regions. Therefore, the marginal resource cost of increasing T_i is proportional to

$$s_i^{TL} - s_i^r \epsilon_i^{r,T} + \left(s_i^{TL} - s_i^r \epsilon_i^{r,L} \right) \frac{\eta \epsilon_i^{V,T}}{1 - \eta \epsilon_i^{V,L}}. \quad (\text{A.78})$$

The first-order condition for maximizing g subject to the transformed budget constraint requires the marginal effect on population growth to be proportional to the marginal resource cost, which gives

$$\Omega_i \propto s_i^{TL} - s_i^r \epsilon_i^{r,T} + \left(s_i^{TL} - s_i^r \epsilon_i^{r,L} \right) \frac{\eta \epsilon_i^{V,T}}{1 - \eta \epsilon_i^{V,L}}. \quad (\text{A.79})$$

This completes the proof. $\square$

A.6 Decomposition of Utility Changes

Notice that for any variable $x > 0$, $\hat{x} \equiv d \log x$. We have

$$\hat{V}_i = \omega_i^C \hat{C}_i + \omega_i^h \hat{h}_i + \omega_i^n \hat{n}_i. \quad (\text{A.80})$$

Therefore, we have

$$\hat{W}_o = \sum_{i=1}^N m_{oi} \hat{V}_i, \quad m_{oi} \equiv \frac{B_i \left(\frac{1}{d_{oi}} V_i \right)^\eta}{\sum_{k=1}^N B_k \left(\frac{1}{d_{ok}} V_k \right)^\eta}. \quad (\text{A.81})$$

Finally, we have

$$\hat{W} = \sum_{o=1}^N s_o^W \hat{W}_o, \quad s_o^W \equiv \frac{t_o W_o}{\sum_{k=1}^N t_k W_k}. \quad (\text{A.82})$$

Then the aggregate welfare changes can be decomposed into

$$\hat{W} = \underbrace{\sum_{o=1}^N \sum_{i=1}^N s_o^W m_{oi} \omega_i^C \hat{C}_i}_{\text{consumption}} + \underbrace{\sum_{o=1}^N \sum_{i=1}^N s_o^W m_{oi} \omega_i^h \hat{h}_i}_{\text{housing}} + \underbrace{\sum_{o=1}^N \sum_{i=1}^N s_o^W m_{oi} \omega_i^n \hat{n}_i}_{\text{fertility}}. \quad (\text{A.83})$$

B Quantification

B.1 Data Construction

This appendix describes the construction of the dataset used in the calibration. The unit of observation is a commuting zone (CZ), defined following [Autor and Dorn \(2013\)](#).

B.1.1 Demographic and Economic Variables

We draw individual-level data from the IPUMS USA 5 percent sample of the 2000 Census ([Ruggles et al., 2025](#)). IPUMS reports geographic identifiers at the Public Use Microdata Area (PUMA) level. We aggregate all variables to the CZ level using the PUMA-to-CZ crosswalks from [Autor and Dorn \(2013\)](#).

Wages. We compute the average annual wage for individuals aged 16–60 within each CZ.

Fertility. We construct a CZ-level fertility rate for women aged 15–45. We use the Census question on the age of the youngest own child in the household. A woman is classified as having had a birth in the past year if the age of her youngest child is zero. The fertility rate is the ratio of such women to all women aged 15–45 in the CZ.

Housing Rent. We compute the average monthly gross rent at the CZ level, restricting the sample to housing units with cash rent.

Per-Capita Housing. We measure per-capita housing as the total number of bedrooms divided by CZ population.

Population. CZ population is the sum of all individuals enumerated in the CZ.

B.1.2 Migration Flows

We obtain county-to-county migration flows from the Census 2000 Special Tabulation on Geographic Mobility.¹ These tabulations record the number of individuals residing in a given county in 2000 who resided in a different county in 1995. We aggregate these flows to the CZ level using the county-to-CZ crosswalk to obtain a CZ-to-CZ migration matrix L_{oi} .

B.1.3 Trade Flows

Bilateral trade flows across CZs are not directly observed. We construct them in three steps. First, we obtain state-to-state commodity flows from the Commodity Flow Survey and compute great-circle distances between state centroids. Second, we estimate a gravity equation at the state level to recover the distance elasticity of trade, following [Ramondo et al. \(2016\)](#). Third, we compute great-circle distances between CZ centroids and apply the estimated gravity coefficients to construct the CZ-to-CZ trade share matrix λ_{in} .

B.1.4 Housing Supply Elasticity

We combine CZ-level housing data from IPUMS 2000 with the MSA-level housing supply elasticity estimates of [Saiz \(2010\)](#). The matching proceeds in three steps. First, we match IPUMS MSA codes to the MSA identifiers in [Saiz \(2010\)](#), supplemented by manual matching for codes that do not link automatically. Second, for the remaining 30–40 unmatched MSAs, we impute the elasticity using the population-weighted average elasticity of the corresponding state. Third, for counties not belonging to any MSA, we assign the national population-weighted average elasticity. We then aggregate county-level elasticities to the CZ level using population weights.

B.1.5 Panel Data Construction (1980–2020)

For the reduced-form regressions reported in Section 4.3, we construct a balanced panel of 741 CZs observed in three waves: 1980, 2000, and 2020 ($N = 2,223$). CZ definitions follow the 1990 CZ boundaries throughout.

¹<https://www.census.gov/data/tables/2000/demo/geographic-mobility/county-to-county-migration-flows.html>.

Geographic Crosswalks. The 1980 and 2000 Census data are mapped to 1990 CZs using the county-group-to-CZ crosswalk from [Autor and Dorn \(2013\)](#). The 2020 American Community Survey (ACS) data are mapped using the crosswalk from [Autor et al. \(2019\)](#). All 741 CZs are observed in all three waves with no missing values.

Number of Bedrooms (Numbedrooms_{jt}). For each CZ j in year t , we compute the average number of bedrooms per owner-occupied dwelling using the bedroom count variable directly available in the Census/ACS. The sample is restricted to owner-occupied units (excluding rentals) to capture the housing quantity margin relevant to the wealth channel.

Home ownership Rate (Own_j). The share of owner-occupied dwellings (as opposed to renter-occupied) in CZ j , computed from the 1980 Census and held time-invariant across all three waves. Using the 1980 value reduces simultaneity concerns arising from contemporaneous housing tenure responses to fertility decisions.

National House Price (hp_t). Following [Moretti \(2013\)](#), we compute the national average price of owner-occupied housing in year t from Census/ACS microdata as the mean self-reported house value across all owner-occupied units in the United States. This variable serves as the “shift” component of the Bartik instrument, capturing aggregate housing demand shocks that are plausibly exogenous to local fertility patterns.

B.2 Calibration

B.2.1 Model Fit

Figure A.1: Parenting Cost (χ_i) that Exactly Fit Fertility Rate (n_i) in the Data



Notes: The fitted line coefficients (standard errors) in panels (a)–(c) are 1.22 (0.40), 0.02 (0.006), and -0.04 (0.007), respectively.